Calibrated Hallucination Risk Estimation Under Domain and Task Shift Using Evidence-Consistent XGBoost

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Abstract

Large language models answer fluently, and some answers are not supported by the evidence. Detectors rely on a large judge, an encoder, or model internals; these options are costly and rarely report calibration under shift. Feature-based detectors are cheaper, but their calibration, length sensitivity, and shift behaviour are seldom measured together, and HaluEval length is almost a label proxy. This study tests whether a simple feature model can estimate risk and how its calibration and ranking change under shift. The model, evidence-consistent XGBoost (EC-XGB), combines 26 base features, eight claim-level NLI aggregates, and a source indicator. It is trained on HaluEval QA and RAGTruth summarization and data-to-text rows with grouped splits and three seeds, and evaluated with a fixed-hyperparameter ablation, source and target Platt calibration, split and class-conditional conformal prediction, SHAP stability, a length-only baseline, an external ensemble, and judge stress tests. The fixed-hp EC-XGB reaches F1 0.9854 and AUROC 0.9986 on HaluEval QA. Lexical overlap and length carry the in-domain score; removing them costs 0.0171 and 0.0066 F1. On MedHallu, the model ranks the paired hallucinated answer above the grounded one in 86.7% of 1,000 pairs (AUROC 0.821), but flags every row and is badly calibrated (ECE 0.493). On a task shift to dialogue and summarization it collapses, with summarization below a length-only model. On RAGTruth QA it reaches AUROC 0.558; target calibration lowers ECE from 0.731 to 0.125 but flags no target rows at 0.5, conformal coverage hides 0.69 positive coverage, and FaithBench coverage falls to 0.32. The GPT judge reaches F1 0.565 on natural QA, but on a length-matched set EC-XGB reaches 0.907 and the judge 0.865. EC-XGB is a domain-specific risk score, not a general-purpose detector.

1. Introduction

A number of researchers have examined how to apply machine learning techniques to assess the likelihood of a response being a hallucination. One such technique is SelfCheckGPT, which uses several generated answers and assesses their similarity to determine if there has been a hallucination [1]. Another technique is Luna, where a smaller evaluation model is used instead of a larger judge model [2]. A third technique is semantic entropy, which identifies meaning-based groupings of similar responses [3]. A hybrid framework combines frozen encoder features with lightweight classifiers and reports strong benchmark scores [4]. Judges carry high computational costs and encoder models are hard to interpret. Two recent reviews have grouped approaches to detecting hallucinations into three categories including feature-based detectors, encoder-based detectors, and combinations of the above [5, 6]. Both reviews identify evaluation and reliability as two major areas that require additional research, and a third review determines that reliability is an area of concern as well [7].

Six recent studies address the same topic of assessing hallucinations. HaluAgent demonstrated that small language models can select appropriate tools and detect multiple forms of hallucinations, but it required multiple tool calls for each answer [8]. Valentin et al. proposed a calibrated multi-score detector with a detailed cost analysis, but the detector relies heavily on large language model (LLM) scoring for its primary source of information [9]. Sundaragiri et al. proposed a system that uses retrieval, natural language inference (NLI) verification, and an XGBoost classifier, and they report only classification metrics [10]. Haq et al. assigned SHAP-LIME attributions to each word, but did not test whether those values remained consistent across words [11]. Deng et al. found that detectors degrade once they leave the domain in which they were trained, and their method focuses on multi-turn dialogue signals [12]. FaithBench shows that modern LLMs still produce hard summarization hallucinations, and it does not assess calibration risks [13].

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The existing literature leaves four issues unaddressed. Calibration is infrequently reported for lightweight detectors, explanation quality is rarely measured, cross-domain behavior is rarely tested, and error patterns are rarely analyzed case by case. The conventional feature-based baseline is also saturated on HaluEval, depends on answer length, and loses calibration under shift. This paper addresses all four issues by developing a black-box model that takes a query q, a reference context c, and a candidate answer a, and outputs $P(\text{hallucinated} \mid q, c, a)$. The model, evidence-consistent XGBoost (EC-XGB), consists of 26 evidence-consistency features in seven groups, eight claim-level NLI aggregates, and a source indicator, classified by a tuned XGBoost model. Platt and isotonic calibration yield the probability, SHAP explanations reveal feature contributions, artifact-control baselines and group-aware splits reduce shortcut solutions, and validation-only calibration quantifies the reliability problem. The claim-level aggregates support per-claim explanations, and the measured transfer gain comes from multi-source training rather than from those aggregates. Transfer limits are established on RAGTruth and FaithBench. The evaluation adds an independent medical corpus (MedHallu), a non-overlapping HaluBench subset, a task-shift test on HaluEval dialogue and summarization, split and class-conditional conformal prediction, a fixed-hyperparameter feature-group ablation, cross-seed SHAP stability, and length-matched and prompt-sensitivity judge stress tests. The added evidence separates ranking from calibration: the model transfers its ranking to medical pairs but not its probability, and it collapses under a task shift.

HaluEval QA provides 10,000 paired correct and hallucinated answers for training and in-domain testing [14]. RAGTruth provides natural responses for adaptation and recalibration [15]. The scope is question answering with a reference context, and FaithBench serves only as an out-of-domain summarization stress test. The fixed-hp m3 model reaches an F1 of 0.9854 and an AUROC of 0.9986 on HaluEval QA. On natural RAGTruth answers, the flag rate falls from 99.9% to 61.3%, and the AUROC rises from 0.475 to 0.582. On an independent medical corpus, the same fixed-hp model ranks the paired hallucinated answer above the grounded one in 86.7% of 1,000 pairs but flags every row; on a task shift to dialogue and summarization it collapses, with summarization ranking below a length-only model. Target calibration on 5,034 labeled rows lowers the RAGTruth QA ECE from 0.731 to 0.125, but it does not change the ranking. On HaluEval, recall stays above 99% at a five percent false-positive rate, and one analysis takes 127.5 ms at about $0.001 per 1,000 predictions.

2. Conventional Method

2.1. Preliminaries

Fluent hallucinations are claims where the amount of support from the evidence is not enough. The detection can be assessed in terms of precision, recall, F1, or area under the receiver operating characteristic curve (AUROC) in an evaluation setting on labeled data. Typical datasets for testing detectors are HaluEval [14] and TruthfulQA [16], and factual accuracy of long-form text generation can be verified by FActScore [17]. In a deployment context, a detector faces tougher evaluation, because it has to contribute to a decision process and to be explainable. Calibration addresses the first problem – the predicted risk has to represent the true risk, and calibration is measured with the use of expected calibration error (ECE) via reliability diagrams [18]. The second one – having the detector's warning, the person needs an explanation of it. For tabular models such information comes from SHapley Additive exPlanations (SHAP) attributions [19]. Both kinds of evaluation are conducted in the current work on the black box approach to the model.

2.2. Baseline Method

The conventional method is a tuned XGBoost classifier [20] over the 26 hand-crafted features. XGBoost handles nonlinear interactions, trains quickly on tabular data, and supports exact SHAP values. Boosted trees remain competitive with tuned neural networks on tabular data. The model adds trees one at a time, and each tree fits the error left by the previous ensemble. The final score passes through a logistic link, so the output is a probability.

Let x_i be the feature vector of row i and $y_i \in \{0, 1\}$ its label. The model sums K trees, $z_i = \sum_{k=1}^{K} f_k(x_i)$. The training objective combines a differentiable loss with a regularizer over the trees,

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \left(\gamma T_k + \frac{1}{2}\lambda \|w_k\|^2\right), \tag{1}$$

where $\phi = \{f_k\}_{k=1}^{K}$ collects the trees, f_k is the k-th tree with T_k leaves of weight w_k, γ is a leaf-count penalty, and λ is an L2 weight penalty. The loss l is the standard logistic loss and the probability comes from the sigmoid of the summed leaf score, $\hat{y}_i = \sigma(z_i) = 1/(1 + e^{-z_i})$.

At boosting step t, a new tree with T leaves is added, and the objective is approximated by a second-order expansion around the current prediction,

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \tfrac{1}{2} h_i f_t^2(x_i) \right] + \gamma T + \tfrac{1}{2} \lambda \sum_{j=1}^{T} w_j^2, \tag{2}$$

where g_i and h_i are the first and second derivatives of the loss at the current prediction. For a fixed tree structure, the optimal weight of leaf j is

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \tag{3}$$

where $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$ are the sums over the rows I_j that reach leaf j. The gain of a split is

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma, \tag{4}$$

with G_L and H_L for the left child and G_R and H_R for the right child. A split is kept only when the gain is positive. This is the criterion used to grow each tree.

Tree depth, learning rate, tree count, and the two subsample rates are tuned on the training split; the selected values are in Section 4. Algorithm 1 lists the eight-step procedure. The search draws 30 configurations and scores each with grouped five-fold cross-validation. The grouping keeps both answers to the same question in one fold. Each seed trains on 14,000 rows, fits Platt scaling on 3,000 validation rows, selects the threshold on validation, and tests once on 3,000 rows. The test rows are never used for tuning or calibration. Step 8 returns the mean over the three seeds, which is why every seeded aggregate table reports a mean.

Algorithm 1 Baseline feature-based XGBoost (training and evaluation)

Require: labeled rows (q_i, c_i, a_i, y_i) and seeds $\mathcal{S} = \{42, 123, 456\}$
Ensure: trained model, calibrator, and decision threshold
1: extract the 26 evidence-consistency features for every row
2: tune XGBoost with a randomized search and grouped 5-fold cross-validation
3: **for** each seed $s \in \mathcal{S}$ **do**
4: train the classifier on the training split
5: fit Platt calibration and select the decision threshold on the validation split
6: evaluate on the fixed test split
7: **end for**
8: **return** the mean metrics over the seeds

This baseline has three limits. The benchmark is saturated, so extra features do not improve F1. The score is sensitive to answer length, and the probability scale breaks under shift [21]. The baseline also lacks claim-level evidence, trains on one corpus, and reports a source-calibrated probability that is not calibrated under target shift. These limits motivate the three changes in Section 3.

3. Proposed Method

3.1. Working Principle

Figure 1 reads from left to right. It takes a question, reference context, and a candidate answer from a black-box LLM, and produces a risk probability and per-feature explanations. The first stage collects the three inputs: the question, the reference context, and the candidate answer. A missing context is allowed and is treated as a separate condition. The second stage extracts 35 features. The seven base groups cover length and style, lexical overlap, entity novelty, NLI, numeric overlap, hedging, and semantic similarity. Eight claim-level NLI aggregates are added to the 26 base features, and a binary source indicator marks whether the answer came from a natural response corpus. The third stage passes the 35 features to the EC-XGB classifier. The classifier returns a raw score, which is a probability over the hallucinated

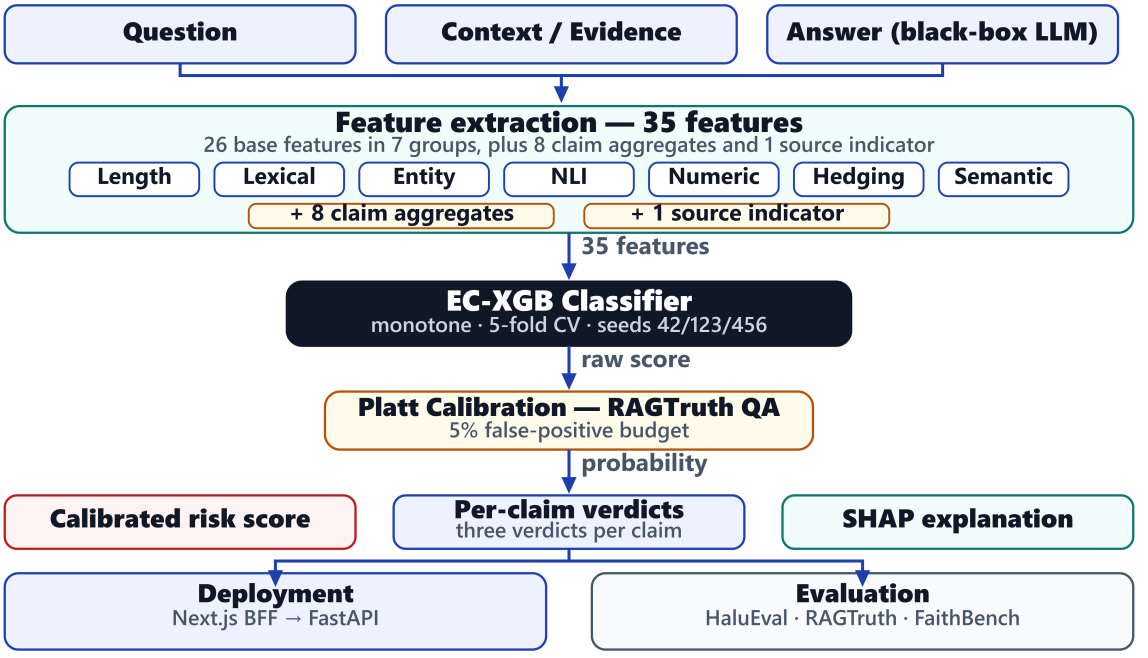


Figure 1: End-to-end pipeline of EC-XGB.

class. The fourth stage applies the Platt calibrator fitted on the RAGTruth QA calibration split. The calibrated score is the number shown to the user. The fifth stage produces the outputs: the risk score, the per-claim verdicts, and the SHAP feature contributions. The lower band shows the deployment path. A Next.js front end calls a FastAPI service, and the service loads the model, calibrator, explainer, and feature models at startup. The rightmost block lists the evaluation corpora. HaluEval measures in-domain performance, RAGTruth summarization and data-to-text measure adaptation to task types seen in training, and RAGTruth QA and FaithBench are held out. The training band records the monotone constraints, the grouped five-fold cross-validation, and the three seeds used for every reported mean.

The task consists of estimating the probability $P(\text{hallucinated} \mid q, c, a)$ of a hallucination, conditioned on a question q, reference context c (if available) and a candidate answer a coming from a black-box LLM. It is not aimed at verifying truthfulness of the candidate answer, but at finding those cases in which the candidate answer does not match the provided evidence, i.e. it does not follow from or contradicts it. The output is a probability, not a binary label. A decision threshold turns the probability into a flag.

3.2. Features

The model uses 26 base features in seven groups (Table 1) and eight aggregate features computed at the claim level (Table 2). For NLI, six signal probabilities come from a DeBERTa-v3 cross-encoder [22] trained on SNLI [23] and MultiNLI [24]. They indicate whether one statement follows logically from another. Semantic similarity uses MiniLM sentence embeddings [25, 26], and entities come from spaCy named entity recognition [27]. Length and style features count characters, words, sentences, and average word length. They are included because the benchmark labels longer answers as hallucinated more often, so length is a strong but fragile signal. Lexical overlap measures how much of the answer appears in the context and in the question. It captures direct copying and is the strongest feature in the SHAP ranking, but it cannot detect a fluent paraphrase that contradicts the evidence. Entity features count named entities and the share that do not appear in the context. A novel entity is a useful warning sign, although a correct answer can also introduce a new name. The NLI features give the probability that the context entails, contradicts, or is neutral to the answer, in both directions. They are the only group that represents logical support rather than surface overlap. Numeric features compare the numbers in the answer with the numbers in the context. A changed number is a common hallucination and is easy to detect. Hedging features count uncertainty words such as “may” or “likely”. They are weak on their own, but they help separate a cautious correct answer from a confident wrong one. Semantic features use cosine similarity between the context and the answer and between the question and the answer. They catch paraphrase and

topic drift that lexical overlap misses. The eight claim-level aggregates in Table 2 repeat the NLI and overlap signals at the level of atomic claims, which is the level a reviewer actually checks. Entailment-based claim metrics are also used by RAGChecker [28].

Table 1
Base feature groups (26 features total).

Group	Features	Source
Length/style	n_chars, n_words, n_sentences, avg_word_len	regex
Lexical	overlap vs. context & question, Jaccard ×2	token sets
Entity	entity counts, overlap ratio, novel-entity ratio	spaCy NER
NLI	entail/contradict/neutral × 2 directions	cross-encoder NLI
Numeric	number counts, overlap ratio, novel numbers	regex
Hedging	hedge count, hedge density	lexicon
Semantic	cosine(context,answer), cosine(question,answer)	MiniLM

Table 2
Claim-level NLI aggregates (8 features).

Signal	Features	Source
Contradiction	max and mean contradiction, contradicted-claim ratio	NLI cross-encoder
Support	supported-claim ratio, unsupported-claim ratio, min and mean entailment	NLI cross-encoder
Decomposition	number of claims	sentence splitter

Note. The answer is split into at most six atomic clauses. Conjunctions such as "and" or "but" are broken into separate clauses. Each clause is scored against its four best-matching context sentences, and the pair probabilities are aggregated.

3.3. Evidence-Consistent XGBoost

EC-XGB alters three key aspects of the conventional XGBoost learner. These changes allow each component of the learning process to be assessed individually. First, raw-length counts are replaced with log-scaled values, and monotonicity constraints ensure that risk cannot decrease as contradiction or novelty rises and cannot increase as entailment, overlap, or semantic similarity rises. Raw-count length is replaced with its log-scaled value, $\ell' = \log(1+\ell)$. Second, the eight claim-level aggregates in Table 2 are added to the evidence signals that the deployed verification layer reports, so the classifier sees the same signals. Consider an answer composed of K claims, with $C(k)$ the contradiction and $E(k)$ the entailment of claim k, and let v_k be the verdict of claim k. The aggregate vector is

$$\psi(x) = [\max_k C(k), \overline{C}, r_{\text{con}}, r_{\text{sup}}, r_{\text{unsup}}, \min_k E(k), \overline{E}, K], \tag{5}$$

$$r_{\text{con}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{contradicted}], \quad r_{\text{sup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{supported}], \tag{6}$$

$$r_{\text{unsup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{unsupported}]. \tag{7}$$

Here $\overline{C}$ and $\overline{E}$ are the means of $C(k)$ and $E(k)$ over the claims. Let $\phi'(x)$ be the complete model input and $\tilde{\phi}(x)$ the remaining 24 base features excluding the two log-scaled length columns. There are 35 input features in total (24 base

+ 2 log-scaled length + 8 claim + 1 source). The claim count K is kept as a feature because an answer with many claims is harder to verify. The third modification is the addition of a source indicator $s \in \{0, 1\}$ as a training feature for RAGTruth summarization and data-to-text tasks. Since all RAGTruth QA samples are filtered out during the training phase, the QA task itself is still zero-shot. The deployed display score is fitted on the labeled target samples and is therefore not zero-shot. This combination of modifications is termed EC-XGB. The deployed checkpoint uses seed 42.

Algorithm 2 lists the training and deployment procedure. Steps 1 and 2 build the feature matrix. The length columns are log-scaled, and the monotone constraints are attached to the features they govern. Steps 3 to 9 repeat the fit for the three seeds. Each repetition builds the 35-feature input, trains on HaluEval plus the RAGTruth non-QA rows, fits the display calibrator on the 5,034 RAGTruth QA calibration rows, selects the source threshold on the HaluEval validation split, and selects the target threshold on the calibration rows at a 5% false-positive budget. The QA rows are excluded from training in every repetition, which keeps the QA test zero-shot. The output is the model, the display calibrator, and the two thresholds.

Algorithm 2 Evidence-consistent XGBoost (EC-XGB) training and deployment

Require: labeled HaluEval QA rows, RAGTruth non-QA rows, seeds $\mathcal{S}$
Ensure: deployed model, display calibrator, source threshold, target threshold
1: extract the 26 base features and the eight claim-level aggregates
2: log-scale the length features and set the monotone constraints
3: **for** each seed $s \in \mathcal{S}$ **do**
4: build the 35-feature input with the source indicator
5: train XGBoost on HaluEval plus RAGTruth non-QA rows, and keep every RAGTruth QA row out of training
6: fit the display calibrator on the RAGTruth QA calibration split (P1)
7: select the source-only threshold on the HaluEval validation split (P0)
8: select the target threshold on the same calibration split at five percent FPR (P2)
9: **end for**
10: **return** the deployed model, calibrator, and thresholds

The contribution is the multi-source training protocol and the evaluation, not a new learner. Each piece is standard on its own, and together they hold the in-domain score, cut the false-alarm rate on the task types seen in training, and produce feature-level attributions whose reliability is measured. The deployed system adds claim-level verification with evidence retrieval and a display calibrator fitted on the target domain. Relative to the closest feature-based detector [10], the claim-level features support per-claim explanations. The measured transfer gain comes from multi-source training and the calibration protocol. Relative to the LLM-scored detector [9] and the tool-based detector [8], the difference is that inference needs no LLM judge calls; the feature pipeline still runs three small pretrained models. Their scores come from other corpora and other protocols, so the comparison is about the protocol and not about the numbers. None of the three reports the calibration error or the shifted flag rate that this study measures.

3.4. Calibration and Explainability

Source calibrators are fit on the validation split only. Platt scaling fits a sigmoid $p(x) = \sigma(Af(x) + B)$ to the raw score $f(x)$, where A and B are fit by maximum likelihood [29]. Isotonic regression fits a non-decreasing step function g so that $p(x) = g(f(x))$ [30]. Both are compared against the raw model on the test split. The comparison uses expected calibration error over $M = 10$ bins,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{8}$$

where $\mathrm{acc}(B_m)$ and $\mathrm{conf}(B_m)$ are the accuracy and mean confidence in bin B_m, together with the Brier score $\mathrm{BS} = \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2$ and reliability diagrams [18]. Post-hoc calibration is not robust when the target distribution moves [31]. A second step targets a new domain. The calibrator can be re-fit on a small labeled sample of the target corpus, and the effect is measured on a disjoint held-out part of the target corpus. The deployed EC-XGB display score follows the same rule: its calibrator is fitted on the 5,034-row RAGTruth QA calibration split, and the source indicator is set to one for natural responses. This target calibration needs labeled target rows and is not zero-shot. The source-only protocol uses the HaluEval validation split for both the calibrator and the threshold.

The global SHAP [19] explanations from a TreeExplainer applied to the raw XGBoost model were used to create Mean Absolute Importance summaries. Local explanations highlighting the contributions of features involved in a particular prediction are given. Post-hoc attribution methods can be unstable even when the model is fixed [32]. Since the feature-attribution techniques might be highly unreliable, the reliability of the explanations has been evaluated directly [33]. The test responses have been perturbed in six different ways (by replacing entity names, numbers, and dates, eliminating support context, inserting additional context, and shuffling clauses). This study quantifies the reliability by calculating four different measures (the mean absolute score change, the fraction of large score changes, the top-1 flip rate, and the Spearman rank correlation between the original and perturbed scores).

3.5. System and Reproducibility

These methods are used in a two-tiered web application. The system is a Next.js frontend and a FastAPI backend that loads the model, calibrator, explainer, and feature models at startup. Three checkpoints appear in Section 5. B2 is the conventional 26-feature model trained on HaluEval only. It supplies the conventional row of the model-comparison table, the conventional rows of the target-calibration table, the length analysis, and the audit. B6 m0 is a separate 26-feature tuning run under the EC-XGB protocol, reported as m0 in the component ablation and as the conventional rows of the cross-domain table. EC-XGB m3 is the 35-feature multi-source model that supplies the remaining EC-XGB rows, the feature-group ablation, and the deployed system. Claims are split from the answer and checked with sentence-level NLI against the provided context, an indexed collection, or live web search. A support-first rule treats any supporting sentence as evidence, and a contradiction counts only when no support exists. A relevance gate removes retrieved sentences with no claim vocabulary, and the verdicts are shown per claim.

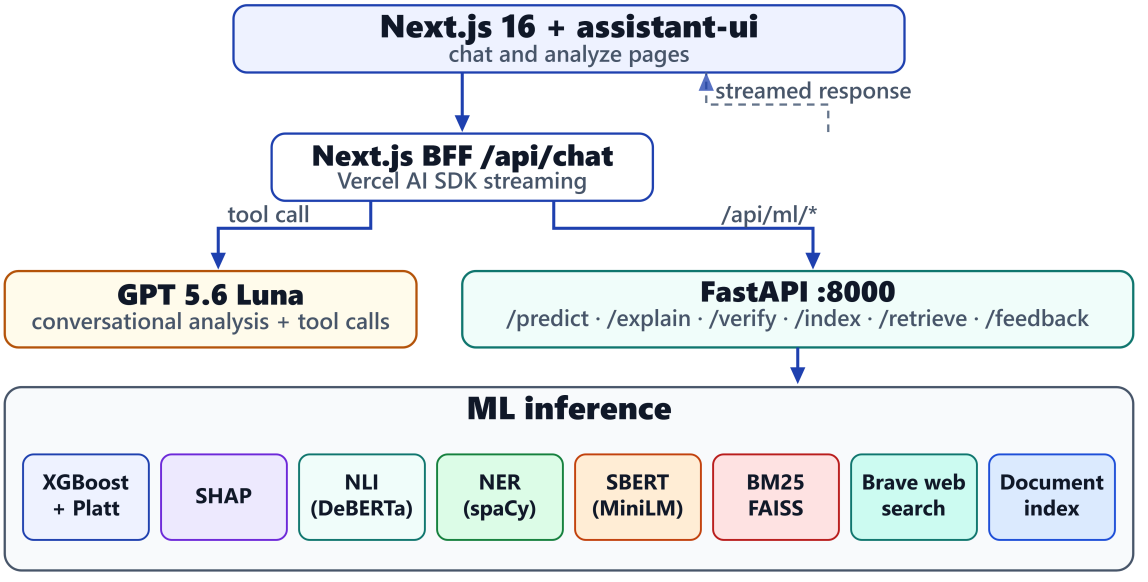


Figure 2: Deployment architecture of the HaluRISC system.

Figure 2 shows the deployment view. The browser calls the Next.js BFF, which streams chat tokens and forwards tool calls to the model. The analyze route calls FastAPI through the `/api/ml/*` rewrite. FastAPI loads EC-XGB, the Platt calibrator, SHAP, NLI, spaCy, and MiniLM at startup. The lower band lists the inference components. A tool call goes to the judge, and a prediction request goes to `/predict`, `/explain`, or `/verify`. Figure 1 shows how one answer is scored, and Figure 2 shows where each component runs.

The verdict card sets high risk for a contradicted claim, medium for an unsupported claim, and low for a fully supported claim. The card's percentage comes from the display calibrator but is adjusted by the per-claim verdict. Figure 3 shows the hallucinated-number case: the context says 5 days per decade, the answer says 10, the raw score is 0.999, and the target-calibrated display score is 35.4%. The number claim is marked unsupported rather than contradicted, so the card reports medium risk. This is a limitation of the support-first rule on numeric claims, not a contradicted-claim example; claim-verdict accuracy is not evaluated here. A 100-claim pre-annotation package

covering 50 answers is released for a three-author blinded review with AI pre-annotation; its AI pre-labels are 63 supported, 14 unsupported, 5 contradicted, and 18 unsure, and they are not reported as human accuracy. The card shows the target-calibrated display score and the risk band; the comparison panel in the application lists the raw score of EC-XGB beside the conventional model and the other baselines. The requirements file pins every dependency, split indices are stored with the artifacts, and the audit scripts check the rebuilt and unchanged reported values against the frozen artifacts. The audits run after every build and fail on any mismatch.

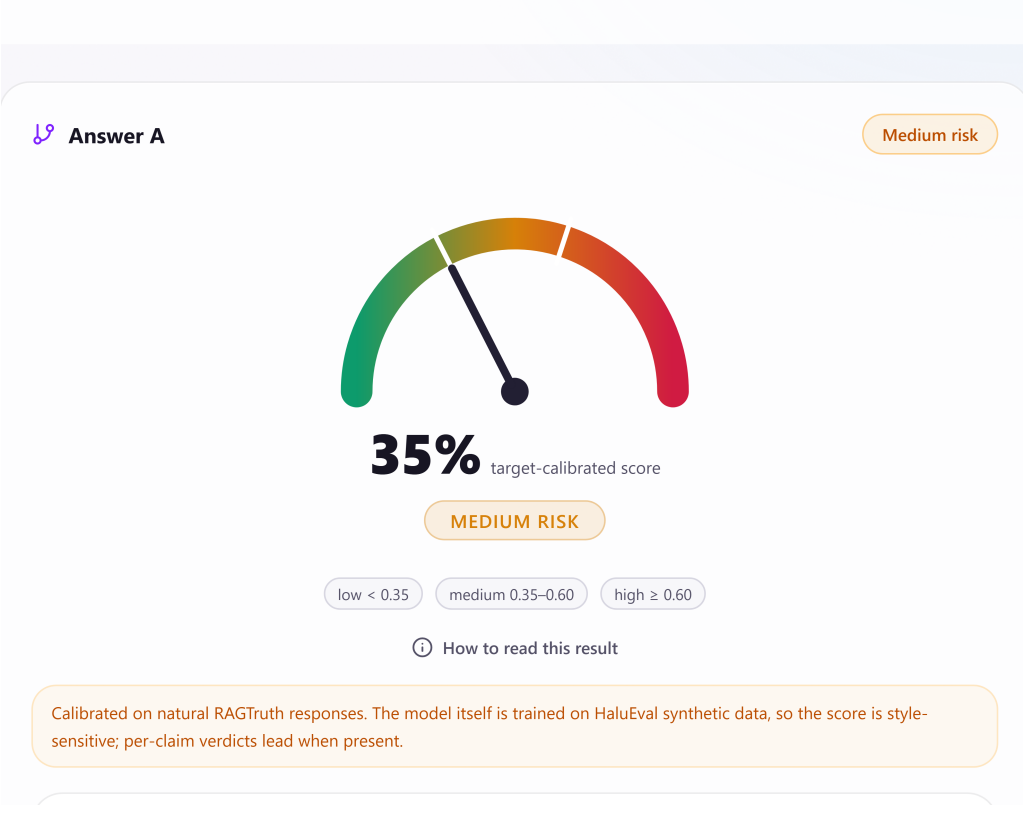


Figure 3: Risk card for the hallucinated-number case, showing the target-calibrated display score.

4. Experiments

4.1. Experimental Setup

All experiments were run on a Windows-based computer (laptop), with an NVIDIA RTX 3060 6 GB GPU, CUDA 12.8, torch 2.11.0+cu128, and 32 GB of RAM. The programming language used to run the code is Python. The Python version used in this experiment is 3.12. The code utilizes the following packages: scikit-learn 1.9.0, XGBoost 3.3.0, SHAP 0.52.0, spaCy 3.8.14, and sentence-transformers 5.6.1. Each run can be reproduced because of saved split indices and centralized seeds. A randomized search was performed over 30 hyperparameter configurations with grouped five-fold cross-validation. For the seed-42 conventional model, the configuration has a maximum tree depth of 5, a learning rate of 0.05, 300 trees, a subsampling ratio of 0.7, and a column sampling ratio of 0.7. Each seed is tuned separately for the main comparison, and the component ablation in Section 5 holds one configuration fixed across variants so that the feature effects are not confounded with tuning. Seeds 42, 123, and 456 are repeated where a seed applies; the single-seed tables are labeled.

4.2. Dataset

The QA subset of HaluEval [14] serves as the training data. This dataset has 10,000 questions. Each question has a knowledge passage that it is based upon, a correct answer, and a hallucinated version of that answer. Therefore, there are 20,000 rows of labeled data. The 20,000 rows are split 70/15/15 at the group level into 14,000 training, 3,000 validation,

and 3,000 test rows. Since both versions of the answer to each question come from the same knowledge passage, the model can easily learn the relationship between the context and the answers, so the split is at the group level. About 85% of HaluEval was generated by ChatGPT through sampling and filtering [14, 34]. The boundary between a factual error and a hallucination is therefore not always clean. Transfer was evaluated using two datasets. First, RAGTruth [15]: a collection of responses produced through natural processes that have been annotated with word-level spans by humans. Second, FaithBench [13]: a hallucination-based summarization benchmark based on outputs from modern large language models. RAGTruth ships official splits. Summarization has 4,758 training and 900 test rows, and data-to-text has 5,298 training and 900 test rows. The QA task contributed 5,934 rows in 989 groups, of which 900 rows in 150 groups form its test set and 5,034 rows form the calibration block. The groups do not overlap. The three 900-row test splits are held out from training, so the pooled RAGTruth test set has 2,700 rows. The positive rate is 50.0% on HaluEval, 34.9% on the pooled RAGTruth test split, 17.8% on the RAGTruth QA test rows, 31.1% on the RAGTruth QA calibration rows, and 68.1% on FaithBench. The summary of the corpora is provided in Table 3.

Three further corpora test external validity without training, calibration, or threshold selection. MedHallu pqa_labeled [35] provides 1,000 medical QA pairs with human-annotated gold answers and paired LLM-written hallucinations, each with gold knowledge, giving 2,000 rows at 50.0% positive. HaluBench [36] supplies 900 rows from DROP, FinanceBench, and covidQA, 300 rows per source at 50.0% positive; its HaluEval, RAGTruth, and PubMedQA rows are excluded to avoid double counting and overlap with MedHallu. Labels are HaluBench PASS/FAIL decisions, not the original source labels, and the DROP and FinanceBench rows were converted to question-answer form by the HaluBench authors. The HaluEval dialogue and summarization subsets provide 1,000 pairs each, 2,000 rows per task at 50.0% positive, as a task-shift test; they share the HaluEval generator but change the task, so they are not a second independent QA corpus.

Table 3
Datasets used for training and evaluation.

Corpus	Rows	Groups	Use
HaluEval QA	20,000	10,000	train / validation / test
RAGTruth QA	5,934	989	held-out test + calibration
RAGTruth full	17,790	2,965	train + pooled test (2,700)
FaithBench	750	750	zero-shot
MedHallu	2,000	1,000 pairs	zero-shot external
HaluBench subset	900	3 sources	zero-shot external
HaluEval dialogue	2,000	1,000 pairs	task-shift test
HaluEval summarization	2,000	1,000 pairs	task-shift test

Note. RAGTruth QA is a subset of RAGTruth full: its 900 test rows and 5,034 calibration rows are already counted there, so the rows are not additive. The pooled RAGTruth test split holds the 900 summarization, 900 data-to-text, and 900 QA test rows. The remaining 15,090 rows form the official training split, and the 5,034 QA rows in it serve as the calibration block. The five zero-shot corpora (FaithBench, MedHallu, HaluBench, and the two task-shift sets) are never used for fitting or threshold selection.

Text is normalized for whitespace. Lexical features work on lowercased token collections. NER and NLI use the original case. Missing context is treated as a special condition. Lexical overlap is 0, and NLI has all probabilities equal to one third. All corpora were developed under their original licenses: HaluEval, RAGTruth, MedHallu, and the HaluEval dialogue and summarization subsets (MIT); FaithBench (CC BY-NC-SA 4.0); and the HaluBench subset (CC BY-NC 2.0). The raw data is not redistributed. However, the repository ships the download scripts, the source revisions, and the SHA-256 hashes, which allows regeneration of each of the splits based upon the original repositories.

4.3. Baselines and Experimental Procedure

Twelve models are tested. The heuristic sets risk to one minus lexical overlap. Logistic regression and a 300-tree random forest are the classical comparators; the conventional XGBoost model [20] uses the 26 base features. Three controls isolate the feature design: NLI-only, answer-only TF-IDF, and full-input TF-IDF. A majority classifier, a fine-tuned RoBERTa-base [37], and a reduced reimplementation of the lightweight ensemble of Sundaragiri et al. [10] complete the set; the two judges are in Section 5.

Internal-state detectors are outside the black-box setting [38]. The RoBERTa baseline receives up to 256 tokens and trains for three epochs. EC-XGB's display calibrator is fitted on the RAGTruth QA split; the in-domain Platt calibrator is fitted on validation. Metrics are reported to four decimals, NLL to two, and rates to one. Significance uses McNemar on seed-42 predictions, 2,000 two-level bootstrap resamples, and Wilcoxon across seeds. No question appears in both training and evaluation. Calibration is compared on raw, Platt, and isotonic scores; Table 4 defines P0, P1, and P2, and every protocol uses the same test rows.

Table 4
The three evaluation protocols.

Protocol	Calibrator fit	Threshold fit	Zero-shot
P0 source-only	HaluEval validation (3,000)	HaluEval validation	yes
P1 target-calibrated	RAGTruth QA calibration (5,034)	fixed at 0.5	no
P2 target-thresholded	RAGTruth QA calibration (5,034)	same rows at 5% FPR	no

Note. Only P0 is zero-shot on the target corpus. P1 and P2 read labels from the target calibration split. The 900-row test split is disjoint from that calibration split in every row.

P0 is source-only; P1 fits the calibrator on the target split and uses 0.5; P2 also selects the threshold on the target split at 5% FPR. The test rows are unseen at calibration time, and the deployed system uses P1; P2 is the thresholded alternative. SHAP [19] TreeExplainers give global and local explanations. Two checks test reliability: the rank agreement between the m3 and B2 SHAP importances and a six-operator perturbation pilot. Content edits should change the score; structural edits should not. Latency is measured on 200 test rows on the reference laptop. The same rows are timed with GPT 5.6 Luna under the LLM-as-a-judge protocol [39], and costs are computed from the recorded token counts.

4.4. Evaluation Metrics

Two families of metrics are reported. Threshold-dependent metrics measure the decision at a fixed cut-off. Threshold-free metrics measure how well a model ranks answers. Both are needed because a model can rank well and still make poor decisions when the threshold is fixed. The confusion matrix supplies the four counts used below, TP, FP, FN, and TN. Precision is the share of flagged answers that are truly hallucinated, and a high value means a reviewer can trust an alert. Recall is the share of hallucinated answers that the detector catches, and a high value means few bad answers pass. F1 is the harmonic mean of the two, and it punishes a model that is strong on one metric and weak on the other. The false-positive rate is the share of correct answers that are flagged. It is the operating constraint under distribution shift, where most natural answers are correct. The flag rate is the share of answers above the decision threshold, and it measures the review workload. A flag rate near 100% means the detector sends every answer to a human.

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F_1 = \frac{2PR}{P+R}, \tag{9}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}, \qquad \mathrm{Flag} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\hat{p}_i \geq \tau]. \tag{10}$$

The second group measures ranking, calibration and paired error patterns. AUROC is the probability that a randomly chosen hallucinated answer receives a higher score than a randomly chosen correct answer. It measures ranking and does not depend on a threshold, and a value of 0.5 is chance. PR-AUC is the area under the precision-recall curve, and it is more sensitive than AUROC when the positive class is rare. RAGTruth QA has a positive rate of 17.8%, so PR-AUC is reported beside AUROC. MCC uses all four cells of the confusion matrix and ranges from -1 to 1, with zero at chance. Accuracy is misleading on an imbalanced test set, so MCC is the safer single-number summary. Negative log-likelihood rewards confident correct probabilities and penalizes confident wrong ones, and lower is better. The Wilson interval gives a 95% range for a proportion and is used for the P2 recall and false-positive rate, where the counts are small.

$$\mathrm{AUROC} = \int_0^1 \mathrm{TPR}(u)\, d\,\mathrm{FPR}(u) = P(s^+ > s^-), \tag{11}$$

$$\text{PR-AUC} = \int_0^1 P(r)\,dr, \qquad \text{MCC} = \frac{\text{TP}\cdot\text{TN} - \text{FP}\cdot\text{FN}}{\sqrt{(\text{TP}+\text{FP})(\text{TP}+\text{FN})(\text{TN}+\text{FP})(\text{TN}+\text{FN})}}, \tag{12}$$

$$\text{NLL} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\right], \tag{13}$$

$$\text{CI}_{95\%} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1+\frac{z^2}{n}}. \tag{14}$$

5. Results

5.1. Model Comparison

Table 5 reports the HaluEval test results for eleven baselines and EC-XGB, averaged over the three seeds where a seed applies. The F1 ranking is tight. Logistic regression is 0.019 behind the conventional model, the random forest is 0.0004 behind, and EC-XGB is 0.0009 ahead. The 0.0009 lead is much smaller than the width of the seed-42 bootstrap interval for the conventional F1, [0.9812, 0.9895], so the ordering is not resolved by the data. The precision and recall columns show why. The random forest gains recall but loses precision, and EC-XGB improves both slightly. These differences are small because the benchmark is saturated. The conventional row in Table 5 is the B2 checkpoint. The m0 row in the component ablation uses the fixed configuration described in Section 5, so the two differ by 0.0003 in F1 and 0.0005 in ECE.

The ranking columns tell the same story. AUROC and PR-AUC place EC-XGB first among the feature-based models, at 0.9984 and 0.9987, with the conventional model and the random forest within 0.0006. RoBERTa is higher at 0.9990 and 0.9993. Logistic regression is lower, at 0.9932 AUROC. MCC is 0.9697 for the conventional model and 0.9715 for EC-XGB. The calibration columns are separate from discrimination. The conventional model has the lowest ECE at 0.0045, but its Brier score is higher than EC-XGB and RoBERTa. The two summaries therefore disagree on which model is better. ECE depends on the bin edges, while the Brier score uses every row. The neural baseline is the strongest model in-domain, at 0.9912 F1 and 0.9990 AUROC. The heuristic reaches 0.9352 F1, the length-only control 0.9457, the NLI-only control 0.6714, the answer-only TF-IDF model 0.9224, and the full-input TF-IDF model 0.5999. The majority baseline scores zero. The NLI-only result shows that the six probabilities carry some signal, but not enough. The gap between the two TF-IDF models shows that adding the question and context to a bag-of-words model hurts. The reduced external ensemble reaches F1 0.9517 and AUROC 0.9850. The fixed-hp m3 model is higher by 0.0337 F1 [0.0262, 0.0419] and 0.0135 AUROC [0.0098, 0.0177] on paired bootstrap draws, and its ECE is 0.0078 against 0.0141. The gap is consistent with the nonlinear learner and the wider feature set, but the baseline is a reduced reimplementation on fewer training rows, so the comparison is descriptive.

The XGBoost margin over the random forest is small, at 0.0004 in F1, and the paired McNemar test on the same seed-42 rows gives $p = 0.502$. The difference is not significant. The margin over logistic regression is larger, at 0.0187 in F1, and the same test gives $p = 1.86 \times 10^{-10}$. The margin over the heuristic is 0.0495 in F1, with $p = 2.28 \times 10^{-30}$. The random forest is therefore the only classical baseline that matches XGBoost on this benchmark. A neural baseline trained solely on the source material outperforms EC-XGB in-domain (F1 0.9912 vs 0.9855), but it is near chance on each RAGTruth task (AUROC 0.458 for summarization and 0.491 for data-to-text), against 0.582 for EC-XGB. The pooled seed-mean AUROC is 0.337; pooling mixes between-task score shifts with within-task ranking. Training an adapted encoder on exactly the same multi-source rows raises it to 0.694 on those tasks, above EC-XGB's 0.582, so the gain on seen tasks is not specific to the feature model. This adaptation does not enable transfer learning, since neither the adapted encoder nor EC-XGB generalizes to the unseen QA task (adapted encoder = 0.490 vs EC-XGB = 0.549), and FaithBench is worse for both (adapted encoder = 0.434 vs original EC-XGB = 0.594). Adaptation improves any model's performance on seen tasks, whereas the feature model retains its relative ordering on unseen tasks.

Figure 4 shows the curves behind Table 5 and the external results. The top row is ROC, and the bottom row is precision-recall. The left column is HaluEval, and the right column is RAGTruth. In the HaluEval panels, the NLI-only and full-input TF-IDF curves are the lowest. The heuristic sits in the middle. The conventional model, EC-XGB, the random forest and logistic regression sit near the top-left corner, and their curves overlap because the benchmark is saturated. In the RAGTruth panels, EC-XGB is above the conventional model on summarization and data-to-text. The two models are close to the diagonal on QA. The curves use seed 42, and the table values are three-seed means.

Table 5
Test-set performance on HaluEval QA, mean over seeds 42/123/456 where a seed applies; deterministic rows are single fits (n = 3,000).

Model	P	R	F1	AUROC	PR-AUC	MCC	ECE	Brier
Heuristic (1−overlap)	0.9473	0.9233	0.9352	0.9132	0.8196	0.8723	0.3541	0.2672
Length-only (2 feats)	0.9454	0.9460	0.9457	0.9759	0.9722	0.8913	—	—
Logistic regression	0.9768	0.9553	0.9660	0.9932	0.9901	0.9329	0.0099	0.0253
Random forest	0.9897	0.9789	0.9842	0.9980	0.9983	0.9687	0.0143	0.0127
NLI-only	0.6479	0.6967	0.6714	0.7177	0.7531	0.3189	0.0678	0.2182
TF-IDF (answer-only)	0.9550	0.8920	0.9224	0.9696	0.9665	0.8519	0.0831	0.0662
TF-IDF (full input)	0.6397	0.5647	0.5999	0.6754	0.6785	0.2484	0.0679	0.2328
External ensemble[†]	0.9508	0.9527	0.9517	0.9850	0.9856	0.9033	0.0141	0.0391
XGBoost (conventional)	0.9932	0.9762	0.9846	0.9979	0.9983	0.9697	0.0045	0.0117
EC-XGB (m3)	0.9939	0.9773	0.9855	0.9984	0.9987	0.9715	0.0074	0.0108
RoBERTa-base	0.9955	0.9869	0.9912	0.9990	0.9993	0.9825	0.0070	0.0081
Majority (negative)	0.0000	0.0000	0.0000	—	—	0.0000	—	—

Note. ECE and Brier are computed on the raw score and measure the reliability of each risk score. The neural baseline is a fine-tuned RoBERTa-base with 256-token inputs. The majority baseline predicts the negative class for every row, so probability metrics are undefined. EC-XGB is the m3 configuration. The conventional row is the B2 checkpoint; the m0 row in Table 6 uses the fixed configuration described there. The length-only row is a single deterministic logistic regression on log-scaled character and word counts, evaluated on the same test split at the 0.5 threshold, and its ECE and Brier were not computed. [†]The external ensemble is a reduced reimplementation of the lightweight feature ensemble of Sundaragiri et al. [10]: five features (ROUGE-L, SBERT cosine, two NLI probabilities, answer length) and a logistic regression, fit on the 14,000 HaluEval training rows, with the threshold selected on validation and evaluated once on test. It is not the original authors' model or reported number.

5.2. EC-XGB and Ablation

Table 6 reports the component ablation under one fixed configuration. All four variants use max depth 5, learning rate 0.05, 300 trees, subsample 0.7 and colsample 0.7, selected once on the m3 seed-42 run. Because the configuration is fixed, the columns measure feature effects rather than tuning effects. The in-domain F1 span is 0.0006: m0 reaches 0.9849, m1 0.9853, m2 0.9849 and m3 0.9854. The claim aggregates change F1 by −0.0004 at m2 relative to m1, inside the two-level interval. The full m3 stack, which adds multi-source training and the source indicator, changes F1 by +0.0001 relative to m1. Neither change is an aggregate gain. The external columns separate the variants more clearly. On the pooled RAGTruth split, AUROC rises from 0.494 for m0 to 0.590 for m3, and raw ECE falls from 0.635 to 0.275. On held-out QA the same comparison is 0.556 against 0.558, which is within seed noise. The McNemar test on the paired seed-42 predictions gives $p = 0.789$ for m0 versus m3, so the ablation does not resolve a difference. The gain on seen tasks comes from the multi-source training rows, not from the claim features. The claim features are retained because they support per-claim explanation, not because they raise the aggregate score. The deployed checkpoint in Table 5 reaches 0.9855 under its own tuned configuration, so the fixed ablation and the deployed model differ by 0.0001 in F1.

Figure 5 shows the seed-level spread behind Table 6. Each point is one seed, and the black bar is the mean. The F1 panel spans 0.9845 to 0.9856 across the four variants, and the three seeds within a variant differ by at most 0.0007. The AUROC panel is tighter, with a within-variant spread below 0.0005. The ECE panel is the noisiest. The m0 variant sits at 0.0047 to 0.0054 with a standard deviation of 0.0004, while m2 and m3 range from 0.0032 to 0.0096 with standard deviations above 0.0016. The figure shows that the mean ordering is m3 > m1 > m2 > m0, but the per-seed ranks flip: seed 42 gives m1 > m3 > m2 > m0, seed 123 gives m1 > m3 > m0 > m2, and seed 456 gives m3 > m2 > m1 > m0. The differences are below seed noise. The ECE sign is stable: m0 is lower than m3 on every seed, by 0.0014 to 0.0048, so the ablation does not establish a calibration advantage for m3.

Table 7 repeats the ablation on the deployed fixed-hyperparameter m3 model, removing one feature group at a time and keeping the claim features and the source indicator. Lexical overlap is the largest contributor, at −0.0171 F1 and −0.0040 AUROC. Length is second, at −0.0066 F1 and −0.0012 AUROC. NLI is third, at −0.0040 F1. Entity, numeric, hedging, semantic, claims and source each change F1 by less than 0.0011. The claim block costs 0.0010 F1,

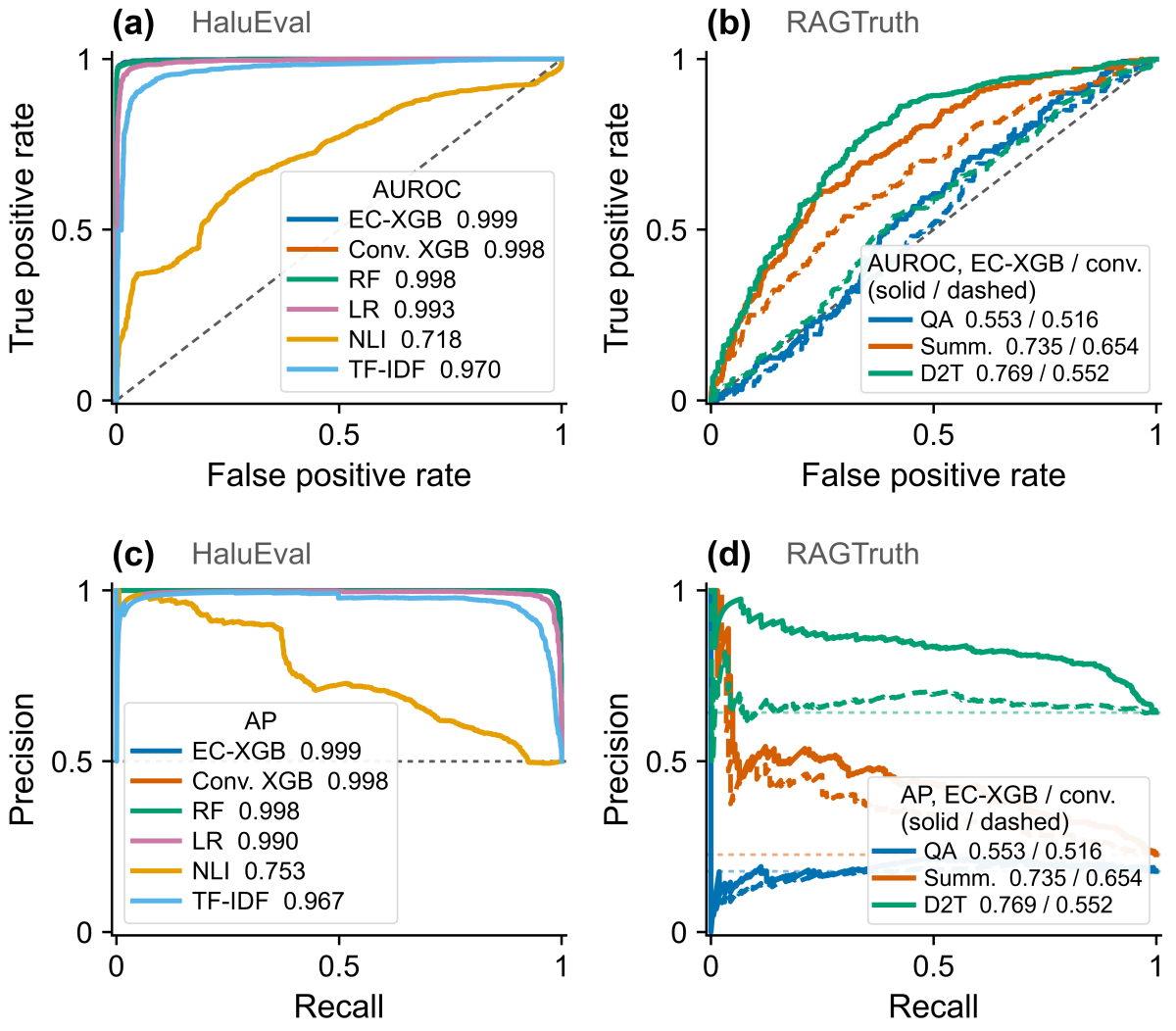


Figure 4: ROC and precision-recall curves on HaluEval and RAGTruth (seed 42).

Table 6
EC-XGB component ablation on the HaluEval test set with one fixed hyperparameter configuration for every variant, mean over three seeds (n = 3,000 for HaluEval columns; n = 2,700 for RAGTruth columns).

Variant	Features	F1	AUROC	MCC	ECE	RAGTruth AUROC	RAGTruth ECE
m0 conventional	26	0.9849	0.9979	0.9701	0.0050	0.4937	0.6352
m1 debias	26	0.9853	0.9976	0.9710	0.0042	0.5567	0.6260
m2 + claims	34	0.9849	0.9985	0.9701	0.0051	0.5461	0.6359
EC-XGB (m3)	35	0.9854	0.9986	0.9713	0.0078	0.5898	0.2746

Note. ECE and Brier are computed on the raw score. All variants use max depth 5, learning rate 0.05, 300 trees, subsample 0.7 and colsample 0.7, selected once on the m3 seed-42 run; no variant is tuned separately. m0 is the conventional reference. m1 replaces raw length counts with log-scaled values and adds monotone constraints (26 features). m2 adds the eight claim aggregates (34 features). m3 adds the RAGTruth non-QA training rows and a source indicator (35 features). The last two columns use the pooled RAGTruth test split (2,700 rows).

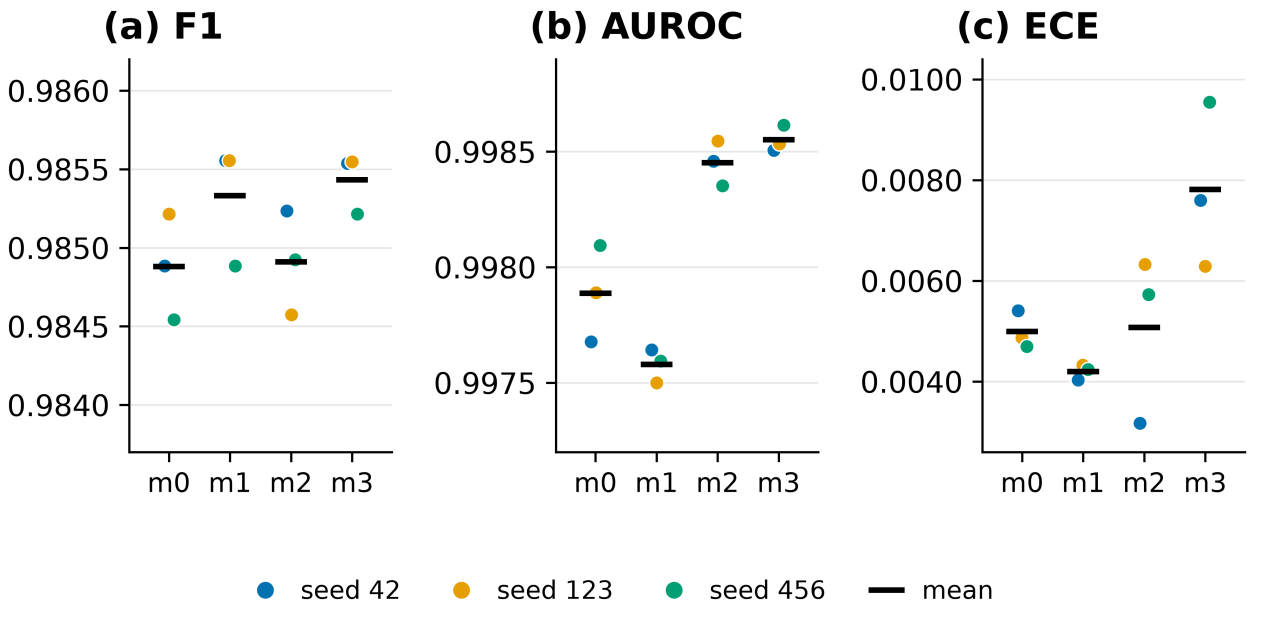


Figure 5: Per-seed F1, AUROC and ECE for the fixed-hyperparameter B6 variants.

which is about the same size as the seed standard deviation of the full model (0.0010), so an aggregate claim gain is not resolved. The ordering matches the conventional ablation: lexical overlap and length carry the in-domain score, and the remaining groups add ranking information without moving F1 far. The m3 reference is refit within this ablation; its F1 is 0.9861 against 0.9854 for the same configuration in Table 6, a 0.0007 difference within the seed spread, and every cell in Table 7 is a delta from that refit.

Table 7
Feature-group ablation of the fixed-hyperparameter EC-XGB m3 model on the HaluEval test set, mean over seeds 42/123/456 (reference F1 0.9861, AUROC 0.9985, MCC 0.9726; n = 3,000).

Removed group	Features	ΔF1	ΔAUROC	ΔMCC
length	4	−0.0066	−0.0012	−0.0127
lexical	4	−0.0171	−0.0040	−0.0343
entity	4	−0.0010	+0.0000	−0.0020
nli	6	−0.0040	−0.0004	−0.0080
numeric	4	−0.0003	+0.0001	−0.0007
hedging	2	+0.0000	+0.0001	0.0000
semantic	2	0.0000	+0.0002	+0.0000
claims	8	−0.0010	−0.0003	−0.0020
source	1	−0.0001	+0.0001	−0.0002

Note. Deltas are relative to the full 35-feature m3 model refit under the same fixed configuration (max depth 5, learning rate 0.05, 300 trees, subsample 0.7, colsample 0.7). Every variant is trained on the same 24,056 rows and evaluated once on the 3,000 test rows. No variant is tuned or early-stopped. The reference differs from Table 6 by 0.0007 F1 because the feature matrix is rebuilt for this run; the difference is within the seed standard deviation of the refit (0.0010).

5.3. Calibration, Shift, and External Validity

The raw, Platt-scaled, and isotonic versions of the conventional model give ECE 0.0045, 0.0091, and 0.0070 on HaluEval, so the calibrated ECEs are not lower in this sample. Figure 6 shows why. On HaluEval, the points at the two ends sit on the diagonal; the middle of the score range holds few distinct values, so the curve has only a few stable bins. The Platt and isotonic curves overlap the raw curve at both ends. The m3 raw score is also well calibrated in-domain, with ECE 0.0078, so there is little room for calibration to improve on HaluEval.

Platt scaling fits a two-parameter logistic map from the raw score s to a probability,

$$\hat{p} = \sigma(as + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{15}$$

where a is the calibration slope and b is the intercept. A slope below one means the raw score is overconfident, and a slope near one means the scale is already usable. The Brier skill score compares the calibrated score with a constant predictor at the evaluation prevalence,

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\mathrm{prior}}}. \tag{16}$$

A positive BSS means the score beats the prior, and a non-positive BSS means it does not.

For the conventional model, the RAGTruth QA histogram in Figure 6 shows almost every row above 0.99, at a mean confidence of 0.998 against an accuracy of 0.178. Source Platt leaves the confidence near 0.99. The m3 score behaves the same way. The fixed-hp m3 raw score is severely overconfident: ECE 0.731, Brier 0.689, and a Cox slope of 0.18 (per-seed point estimates 0.16 to 0.23) on the 5,034 calibration rows. A slope of one would be calibrated, so the raw score moves about five to six times too far from 0.5. Source Platt, fitted on HaluEval validation, cannot repair a base-rate shift; its ECE is 0.765 and its Brier is 0.742. Target Platt, fitted on the calibration rows, lowers ECE to 0.125 and Brier to 0.161, but the Brier skill score against the test prior is −0.10. A constant predictor at the test prevalence has Brier 0.146, lower than every calibrated row, and it flags no rows at 0.5. Calibration fixes the scale, not the ranking; AUROC stays at 0.558 before and after. The conventional target-Platt row behaves the same way, at ECE 0.134 and Brier 0.164. The label budget is the practical constraint, and it is a distribution rather than a point.

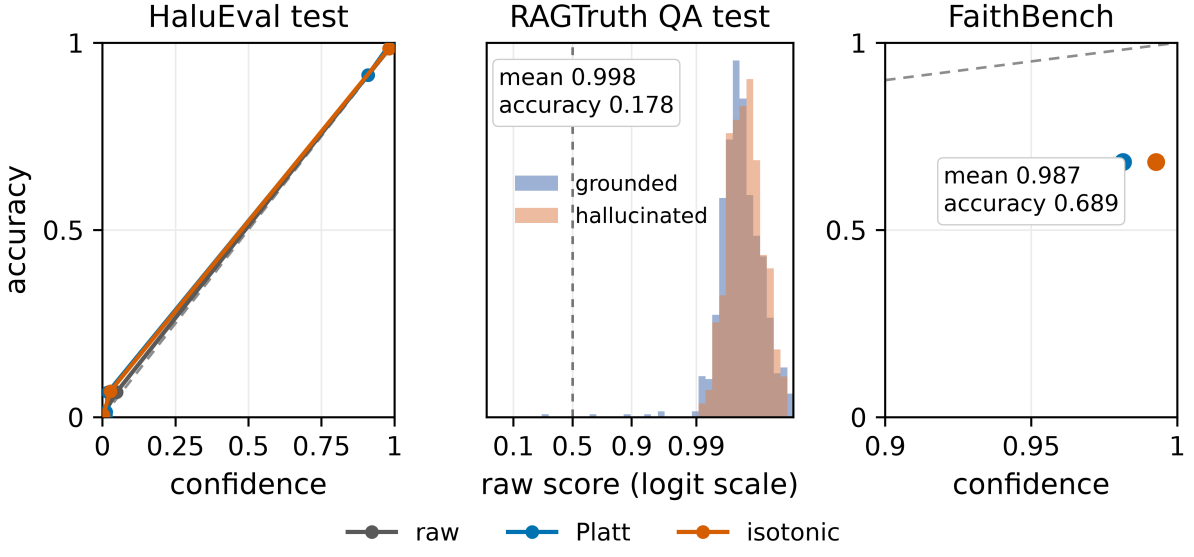


Figure 6: Calibration of the B2 conventional model. Left and right: quantile-bin reliability diagrams for HaluEval and FaithBench (FaithBench is aggregated because its scores concentrate near 1.0). Middle: RAGTruth QA raw-score distribution on a logit scale; both classes pile near 1.0.

The source-only operating point (P0) selects the threshold on the HaluEval validation split at a 5% false-positive budget. On the HaluEval test split, recall there is 99.0–99.5% and the false-positive rate stays near 5%. On RAGTruth QA the same source threshold flags 97.4% of the tuned checkpoint, so P0 is not a usable target operating point. Table 8 reports both models on the same 900-row RAGTruth QA test set, whose prevalence is 17.8%. The calibration split has 31.1% positives. A prevalence-only predictor set to the test prevalence, 0.1778, shows what a constant score can achieve. On the test rows it reaches ECE 0.000 and Brier 0.146, and it predicts no positives at the 0.5 threshold. At the calibration prevalence of 0.3107 the same predictor gives ECE 0.133 and Brier 0.164, which matches the conventional target-Platt row. The calibrator therefore sets the level to the calibration prevalence, and the residual ECE is dominated

Table 8
Calibration of the fixed-hp m3 model and the conventional target-Platt reference on the RAGTruth QA test set (n = 900, 17.8% positive).

Predictor	Fit	ECE	Brier	BSS	Slope	Flagged
Prior (test prevalence)	constant 0.1778	0.000	0.146	0.000	—	0.0%
Prior (cal prevalence)	constant 0.3107	0.133	0.164	−0.12	—	0.0%
m3 raw	none	0.731	0.689	−3.71	0.18 [0.12, 0.27]	96.8%
m3 source Platt	HaluEval val	0.765	0.742	−4.07	—	96.8%
m3 target Platt	RAGTruth cal	0.125	0.161	−0.10	—	0.0%
Conventional target Platt	RAGTruth cal	0.134	0.164	−0.12	—	0.0%

Note. ECE and Brier are on the 900 test rows. BSS uses the test-prevalence prior as the reference. The slope is the Cox slope of the raw score on the 5,034 calibration rows; a value of one is calibrated, and the brackets are the envelope of the three per-seed 95% bootstrap CIs. Row-bootstrap 95% CIs for the headline rows are raw ECE [0.706, 0.757], raw Brier [0.664, 0.713], target ECE [0.099, 0.151], and target Brier [0.151, 0.171]. Platt fits use validation or calibration rows only. The test prior row is zero by construction; the calibration-prior row is evaluated on the test rows. The conventional row uses the B2 target-Platt checkpoint and is shown as the reference behind the P2 analysis.

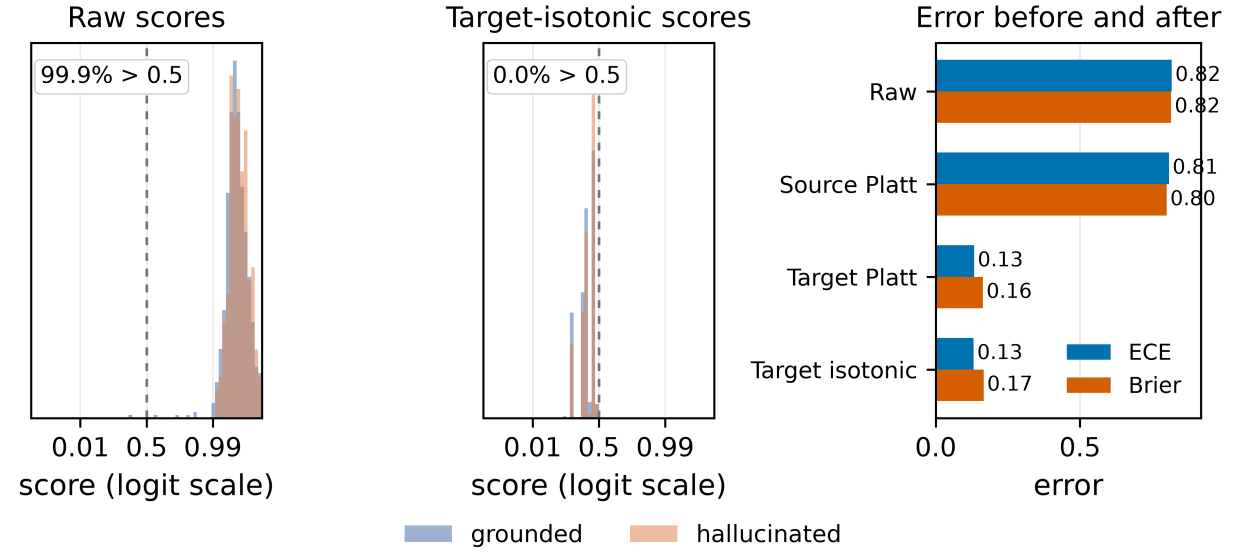


Figure 7: Target-domain recalibration of the conventional model on RAGTruth QA. Left and middle: raw and target-isotonic score distributions on a logit scale. Right: ECE and Brier for raw, source Platt, target Platt, and target isotonic.

by the prevalence gap. At the test prevalence of 0.1778 a constant predictor has Brier 0.146, lower than the target-calibrated rows (0.161–0.164). Figure 7 shows the raw scores and the target-recalibrated scores for the conventional model. Calibration fixes the scale, not the ranking. The conventional target-thresholded mode (P2) selects a threshold at a 5% false-positive budget (0.313) on the calibration split and reaches a test recall of 5.0% at a false-positive rate of 5.8% (precision 15.7%, F1 0.076, Wilson 95% intervals: recall [2.6, 9.6], FPR [4.3, 7.7]). The test rate exceeds the 5% target because the threshold is fixed on calibration. At 17.8% prevalence, random flagging at a 5.8% false-positive rate yields 5.8% recall, so the P2 result is chance-level. At the 0.5 threshold, P1 flags no RAGTruth QA rows, and the maximum target-Platt score (0.313) is below the API's lowest risk band (0.35); the deployed card therefore relies on the per-claim verdict adjustment, not on the pre-verdict score alone. All calibration metrics in this section refer to the pre-verdict display score. The card value is adjusted by claim verdicts and is not covered by these ECEs; the displayed percentage is a design aid, not a validated probability.

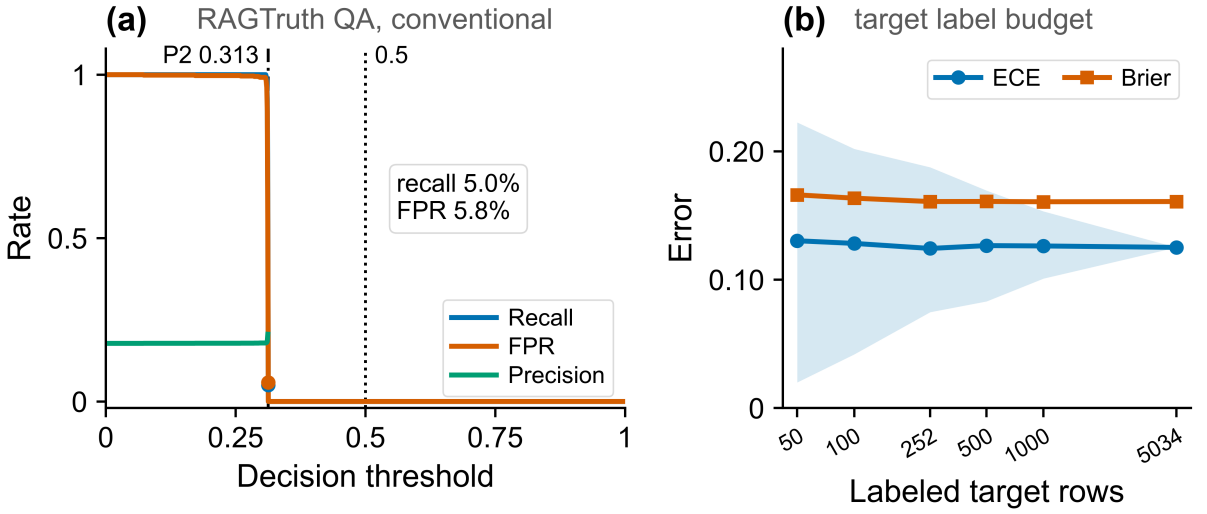


Figure 8: Operating point for the conventional target-Platt model and 100-draw label budget for the fixed-hp m3 calibrator on RAGTruth QA.

Figure 8 shows the operating point and the label budget. Panel (a) sweeps the threshold for the conventional target-Platt model on RAGTruth QA, and the vertical line marks the selected threshold at 0.313, which is the P2 row in Table 8. A lower threshold raises recall and raises the false-positive rate. Panel (b) plots 100 draws at each budget. The mean ECE is 0.130 at 50 rows, 0.124 at 252 rows, and 0.125 at 5,034 rows, but the 95% band narrows from [0.020, 0.223] at 50 rows to [0.075, 0.188] at 252 rows and collapses at the full split. A single 252-row draw can therefore miss the full-split ECE by about 0.06, so the budget saturates in the mean but not in a single draw. A small labeled target sample sets the scale only when its draw is checked against a held-out split.

Table 9 reports the coverage. Split conformal prediction adds a coverage guarantee to the score. The nonconformity score for label y is $s(x, y) = 1 - \hat{p}_y(x)$, where $\hat{p}_1 = \hat{p}$ and $\hat{p}_0 = 1 - \hat{p}$. The calibration quantile and the prediction set are

$$\hat{q} = \text{Quantile}_{1-\alpha}\left(\{s(x_i, y_i)\}_{i=1}^{n}\right), \qquad C(x) = \{y : s(x, y) \le \hat{q}\}. \tag{17}$$

A class-conditional version replaces the single quantile with one quantile per label,

$$\hat{q}_y = \text{Quantile}_{1-\alpha}\left(\{s(x_i, y_i) : y_i = y\}\right), \qquad C(x) = \{y : s(x, y) \le \hat{q}_y\}. \tag{18}$$

The main run fits the Platt map and the quantile on the same 5,034 rows, so its calibration coverage is a construction check. A disjoint sensitivity fits Platt on 2,514 rows and the quantile on 2,520 rows. On that split, the 900 QA test rows reach coverage 0.944 at 90% and 0.972 at 95%, so the marginal guarantee holds with slight over-coverage. The guarantee hides the class split: positive coverage is 0.685 at 90% and 0.842 at 95%, while negative coverage is 1.000,

and the average set size is 1.59 and 1.75. A class-conditional quantile, fitted separately for each label, raises positive coverage to 0.887 at 90% and 0.925 at 95% on QA, still slightly below nominal, at average set sizes of 1.69 and 1.80. The shift is harsher. FaithBench coverage falls to 0.320 at 90% and 0.323 at 95%, and positive coverage is 0.003, because a Platt map fitted on QA sends almost every FaithBench score to the negative side. Conformal prediction therefore certifies the marginal rate only when the calibrator and the quantile are fitted on disjoint rows, and it does not transfer across the task shift. The per-class rate is the one a reviewer needs when positives are 17.8% of the target.

Table 9
Split and class-conditional conformal coverage on the 900 RAGTruth QA test rows and 750 FaithBench rows, mean over seeds 42/123/456. The Platt map and the conformal quantile are fitted on disjoint halves of the 5,034 QA calibration rows.

Dataset	Method	α	Coverage	Pos.	Neg.	Set size
QA test	split	0.10	0.944	0.685	1.000	1.59
QA test	split	0.05	0.972	0.842	1.000	1.75
QA test	class-cond.	0.10	0.886	0.887	0.886	1.69
QA test	class-cond.	0.05	0.936	0.925	0.938	1.80
FaithBench	split	0.10	0.320	0.003	1.000	1.00
FaithBench	split	0.05	0.323	0.006	1.000	1.01

Note. Coverage is the share of rows whose true label is in the prediction set. Pos. and Neg. are the same rate computed within each class. Set size is the mean number of labels in the set, with a maximum of two. The class-conditional rows use a separate quantile per label on QA. FaithBench is a zero-shot summarization corpus and is never used for fitting. The in-sample run, which fits Platt and the quantile on the same rows, gives 0.940 and 0.967 coverage on the QA test rows and 0.900 and 0.950 on the calibration rows; the latter is a construction check and is not reported as the primary result.

Multicalibration by answer length and claim count shows where the average ECE hides error. On the 900 QA test rows, the target-Platt score has overall ECE 0.125. The worst length bin is 60 words or fewer (n=154), with ECE 0.275, and the 61–80 and 81–100 bins are 0.224 and 0.172; the 161+ bin is 0.082. The worst claim-count bin is a single claim (n=36), with ECE 0.346, against 0.086 for six or more claims (n=706). The Brier score does not follow the ECE ranking: it is lower in the short and one-claim bins, so the two summaries disagree about where the calibration error is concentrated. The average is therefore not a safe summary for short answers or one-claim answers, which are the cases where a reviewer most needs a reliable number. FaithBench is not improved by the QA calibrator: its target-Platt ECE is 0.629 against 0.453 for the raw score, so the calibrator is domain-specific.

The conventional reference m0 is trained only on HaluEval, whereas EC-XGB is exposed to RAGTruth summarization and data-to-text examples as well, so the performance of EC-XGB on these two tasks measures adaptation to task types seen in training, not general robustness to distribution shift (Table 10). On the pooled RAGTruth test split, the flag rate falls from 99.9% to 61.3% and AUROC rises from 0.475 to 0.582. The task-specific split reveals the source of improvement. On summarization, AUROC rises from 0.634 to 0.736 and the flag rate falls from 99.8% to 11.6%, so F1 drops from 0.370 to 0.343 as the fixed threshold stops flagging nearly every row. Data-to-text AUROC rises from 0.567 to 0.771. Held-out QA remains near chance (AUROC 0.547 to 0.549; PR-AUC 0.191 for m0 and 0.188 for m3, so the two metrics disagree on the sign of the small difference). FaithBench AUROC rises from 0.509 to 0.594, but F1 falls from 0.813 to 0.120 and raw ECE rises from 0.292 to 0.450, so the raw scores transfer poorly to that corpus. The conventional F1 in Table 10 matches the all-positive baseline to within 0.003 on QA (0.302), pooled (0.518), and FaithBench (0.810) because it flags almost every row.

Table 11 reports the three external corpora and the task-shift pair. MedHallu is the strongest evidence that the ranking survives a domain change and the clearest evidence that the probability does not. The fixed-hp m3 model ranks the paired hallucinated answer above the grounded answer in 86.7% of 1,000 medical pairs [84.5, 88.7], with AUROC 0.821 [0.806, 0.837] on the 2,000 rows. The pair accuracy over M pairs is

$$\mathrm{PA} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\left[\hat{p}(a_m^+) > \hat{p}(a_m^-)\right], \tag{19}$$

where a_m^+ is the hallucinated answer and a_m^- the grounded answer. The conventional m0 model reaches 82.2% and 0.781. The length-only baseline is inverted, at 20.0% and 0.260, because grounded MedHallu answers are longer than

Table 10
External and adaptation performance of B6 m0 and EC-XGB m3 by dataset and task on the held-out RAGTruth test rows and FaithBench, mean over three seeds.

Dataset / task	Model	F1	AUROC	PR-AUC	Flagged	ECE
RAGTruth pooled	Conventional (m0)	0.5181	0.4753	0.3139	99.9%	0.6346
RAGTruth pooled	EC-XGB (m3)	0.5608	0.5815	0.3588	61.3%	0.2782
RAGTruth QA	Conventional (m0)	0.3022	0.5465	0.1911	99.9%	0.8149
RAGTruth QA	EC-XGB (m3)	0.3081	0.5491	0.1884	97.4%	0.7433
RAGTruth summarization	Conventional (m0)	0.3702	0.6343	0.3536	99.8%	0.7499
RAGTruth summarization	EC-XGB (m3)	0.3434	0.7358	0.4449	11.6%	0.0519
RAGTruth data-to-text	Conventional (m0)	0.7830	0.5665	0.6831	100.0%	0.3390
RAGTruth data-to-text	EC-XGB (m3)	0.8234	0.7705	0.8374	74.9%	0.0527
FaithBench	Conventional (m0)	0.8130	0.5089	0.6735	99.5%	0.2917
FaithBench	EC-XGB (m3)	0.1201	0.5941	0.7109	7.3%	0.4500

Note. ECE is computed on raw scores, not on the target-calibrated display score. The conventional rows are B6 m0, a separate 26-feature tuning run under the EC-XGB protocol, and the EC-XGB rows are m3. Every row is a three-seed mean. m0 is trained on HaluEval only, so its external rows are zero-shot. EC-XGB adds the RAGTruth summarization and data-to-text training rows, so those two pairs measure adaptation, and RAGTruth QA is held out. F1 uses the fixed 0.5 threshold, and Flagged is the share of rows above it. RAGTruth pooled uses the 2,700 held-out test rows, each RAGTruth task uses 900, and FaithBench uses 750. The m0 F1 matches the all-positive baseline to within 0.003 on QA (0.302), pooled (0.518), and FaithBench (0.810) because it flags almost every row. These rows use the frozen tuned checkpoint; the fixed-hp judge rows give 96.8% and AUROC 0.558 on QA, within 0.01 of the tuned values.

Table 11
External validity on corpora that are never used for training, calibration, threshold selection, or model choice (scores averaged over seeds 42/123/456). Pair accuracy is the share of the 1,000 MedHallu pairs in which the hallucinated answer scores above the grounded one.

Dataset (n)	Model	F1	AUROC	ECE	Flagged	Pair acc.
MedHallu (2,000)	EC-XGB (m3)	0.667	0.821	0.493	100.0%	0.867
	Conventional (m0)	0.667	0.781	0.499	100.0%	0.822
	Length-only	0.667	0.260	0.500	100.0%	0.200
HaluBench (900)	EC-XGB (m3)	0.588	0.600	0.334	50.3%	—
	Conventional (m0)	0.585	0.609	0.346	51.6%	—
	Length-only	0.441	0.514	0.420	33.2%	—
HaluEval dialogue (2,000)	EC-XGB (m3)	0.670	0.763	0.488	99.3%	—
	Length-only	0.680	0.724	0.438	95.9%	—
HaluEval summarization (2,000)	EC-XGB (m3)	0.583	0.717	0.113	37.5%	—
	Length-only	0.667	0.788	0.500	100.0%	—

Note. MedHallu is medical QA; HaluBench is a non-overlapping subset from DROP, FinanceBench, and covidQA (300 rows each); HaluEval dialogue and summarization are task-shift tests within the HaluEval family, not a second independent QA corpus. The length-only baseline is a logistic regression on log answer length, fit once on HaluEval train only. HaluBench rows are not paired, so pair accuracy is undefined. All evaluation sets are balanced 50/50 by construction, so F1 and flag rate are prevalence-dependent and do not describe natural deployment. Flagged is the share of rows above 0.5. The feature models use the fixed-hp configuration; no external row is used for tuning.

the paired hallucinations (39.7 vs 27.8 words), the opposite of HaluEval (2.2 vs 11.0). The ranking therefore transfers, but the scale does not: m3 flags every row, so its F1 is the all-positive 0.667 and its ECE is 0.493. The source indicator is one for MedHallu and zero for HaluBench; a source-zero sensitivity on MedHallu changes pair accuracy by at most 0.003 and AUROC by at most 0.002, so the indicator does not drive the result. HaluBench is a 30% stratified sample of the 3,000 non-overlapping rows. The overall AUROC is 0.600, with F1 0.588, ECE 0.334, and a 50.3% flag rate. The per-source split is not uniform: covidQA reaches AUROC 0.833 and F1 0.641, while DROP is 0.513 and FinanceBench 0.527. The model transfers to the biomedical subset but is at chance on the discrete-reasoning and finance subsets; the length-only baseline is at 0.514 overall, so 0.600 is not a length artifact.

The task-shift test is the weakest. On 2,000 dialogue rows, m3 reaches F1 0.670 and AUROC 0.763 but flags 99.3% of rows, so its F1 equals the all-positive baseline and its ECE is 0.488; the length-only baseline reaches AUROC 0.724. On 2,000 summarization rows, m3 reaches F1 0.583, AUROC 0.717, ECE 0.113, and flags 37.5%; the length-only baseline reaches AUROC 0.788 and flags every row. The conventional m0 model flags 99.2% of dialogue rows and 97.7% of summarization rows, with AUROC 0.741 and 0.605. Both models fail to transfer, and neither is a cross-task detector. The dialogue result is the all-positive baseline; the summarization result is below the length-only ranking. This is a task-shift test, not a second independent QA corpus.

An index stress test asks whether the weak held-out QA result is a retrieval problem. The hybrid BM25 and FAISS index covers the 989 RAGTruth QA source documents. On a 300-row, 50-group sample, recall@1 is 0.94, recall@5 and recall@10 are 1.00, and MRR is 0.959. Replacing the gold context with the top-5 retrieved passages changes AUROC from 0.545 to 0.521, F1 from 0.345 to 0.349, and ECE from 0.706 to 0.704. The AUROC delta is −0.023 [−0.061, 0.011], which includes zero. Retrieval is therefore not the bottleneck: the model is near chance on RAGTruth QA even with the gold source passage. The qrels are the known source passage, not human web-search judgments, so this measures index recall, not end-to-end search quality. The sample has 50 source groups, so the interval is wide.

Figure 9 shows the raw score distributions of the conventional model on HaluEval and on the external test rows. The first panel is the HaluEval test set, where the two classes separate into two clusters at the ends of the range. The remaining panels are the held-out RAGTruth test rows (900 per task, 2,700 pooled) and FaithBench. In each external panel, both classes pile against 1.0, so the raw score no longer separates hallucinated from correct answers. The 0.5 threshold therefore flags almost every row. This is the visual form of the 99.9% flag rate in Table 10.

5.4. Statistical Analysis

Statistical tests use the paired fixed-hyperparameter predictions and 2,000 two-level bootstrap draws, where groups are resampled within each seed and then seeds are resampled. McNemar's test compares the discordant counts b and c on the paired seed-42 rows, where b is the number of rows where the first model is right and the second is wrong, and c is the reverse. The test statistic is

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \tag{20}$$

with a continuity correction. Holm's method orders the m p-values and adjusts the i-th value as

$$p_{(i)}^{\text{Holm}} = \min_{j \geq i} \left\{ \min \left(1, (m - j + 1) p_{(j)} \right) \right\}, \tag{21}$$

which controls the family-wise error rate across the four baseline comparisons. The paired bootstrap delta is

$$\Delta_b = \theta_{\text{EC}}(\mathcal{D}_b) - \theta_{\text{conv}}(\mathcal{D}_b), \tag{22}$$

where $\mathcal{D}_b$ is a resample of groups and θ is the metric. The 95% interval is the 2.5th and 97.5th percentiles of the Δ_b,

$$\text{CI}_{95\%} = \left[\Delta_{(\alpha/2)}, \Delta_{(1-\alpha/2)} \right]. \tag{23}$$

The two-level version resamples groups within each seed and then resamples seeds,

$$\Delta^* = \frac{1}{S} \sum_{s=1}^{S} \left[\theta_{\text{EC}}^*(\mathcal{D}_s^*) - \theta_{\text{conv}}^*(\mathcal{D}_s^*) \right], \tag{24}$$

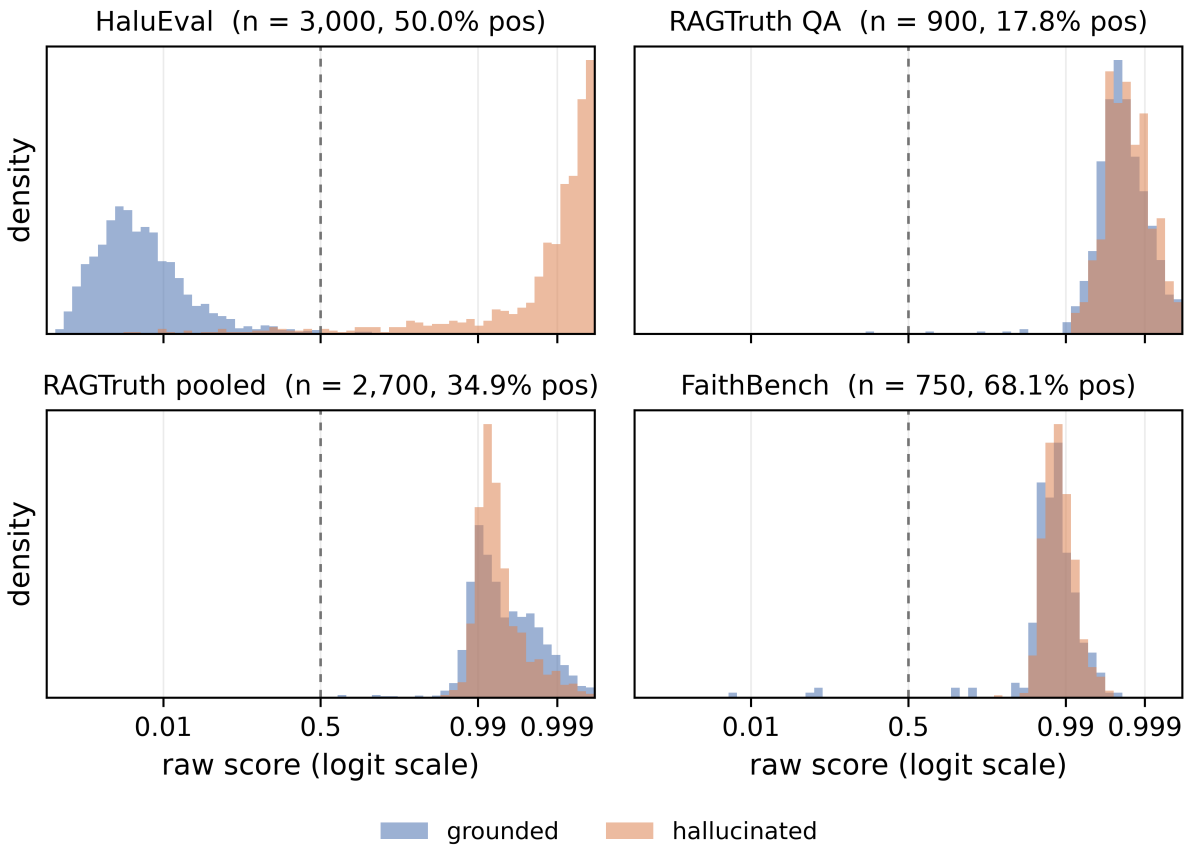


Figure 9: Raw score distributions of the conventional B2 model on HaluEval and the held-out external test rows (RAGTruth QA, pooled RAGTruth, FaithBench), on a logit scale. Both classes pile near 1.0 outside HaluEval.

where D_s^* is a group resample for seed s and $S = 3$. The Wilcoxon signed-rank statistic is

$$W = \sum_{s=1}^{S} \text{rank}(|d_s|)\mathbf{1}[d_s > 0], \tag{25}$$

where d_s is the difference between the two models on seed s and $S = 3$.

Table 12 gives the McNemar counts. EC-XGB and the conventional model are not significantly different under the deployed checkpoints ($p = 0.45$), and the fixed-hyperparameter ablation gives $p = 0.789$ on the same test rows. That is the primary comparison, and it is not part of the Holm family, which covers the four baseline comparisons against the conventional model. At $p = 0.45$ it would not survive correction. The random forest matches the conventional model ($p = 0.502$). Logistic regression and the heuristic are significantly worse ($p = 1.86 \times 10^{-10}$ and $p = 2.28 \times 10^{-30}$), and both survive Holm correction. The Wilcoxon test across three seeds cannot reach significance (minimum $p = 0.25$). Table 13 reports the two-level bootstrap. On HaluEval, the m3−m0 F1 delta is +0.0006 [−0.0010, 0.0021], which includes zero, and the AUROC delta is +0.0007 [0.0002, 0.0012], which excludes zero by a small margin. The m2−m1 F1 delta is −0.0004 [−0.0020, 0.0012], and the m3−m1 delta is +0.0001 [−0.0015, 0.0018], so the claim features do not add an aggregate gain. Their AUROC deltas are +0.0009 and +0.0010, both with intervals above zero, so the added features reorder a few rows without changing F1. On held-out RAGTruth QA the m3−m0 F1 delta is +0.0076 [0.0045, 0.0109], but the AUROC delta is +0.0017 [−0.0270, 0.0291], which includes zero. The F1 gain comes from flagging almost every row, not from better ranking. On the pooled seen tasks the F1 delta is +0.0451 [0.0348, 0.0554] and the AUROC delta is +0.0962 [0.0644, 0.1295], which is the adaptation gain. The paired fixed-minus-tuned contrasts all include zero, so the rebuild does not change these conclusions.

Table 12
McNemar tests on the paired seed-42 HaluEval predictions.

Comparison	b	c	p	Holm p
EC-XGB vs conventional	10	6	0.453	—
EC-XGB vs conventional (fixed hp)	6	8	0.789	—
Conventional vs random forest	12	8	0.502	0.502
Conventional vs logistic regression	69	11	1.86×10^{-10}	3.7×10^{-10}
Conventional vs heuristic	158	9	2.28×10^{-30}	6.8×10^{-30}
Conventional vs NLI-only	1006	26	5.6×10^{-204}	2.2×10^{-203}

Note. b is the number of rows where the first model is right and the second is wrong. c is the reverse. The p values use the chi-square form with continuity correction. Holm values cover the four baseline comparisons against the conventional model. The fixed-hp row compares m0 and m3 under the single configuration of Table 6.

Table 13
Two-level bootstrap deltas for the fixed-hyperparameter ablation (2,000 draws; groups within seed, then seeds). Intervals are 95%.

Dataset	Comparison	ΔF1 [95% CI]	ΔAUROC [95% CI]
HaluEval	m3−m0	+0.0006 [−0.0010, 0.0021]	+0.0007 [0.0002, 0.0012]
HaluEval	m2−m1	−0.0004 [−0.0020, 0.0012]	+0.0009 [0.0004, 0.0014]
HaluEval	m3−m1	+0.0001 [−0.0015, 0.0018]	+0.0010 [0.0004, 0.0016]
RAGTruth QA	m3−m0	+0.0076 [0.0045, 0.0109]	+0.0017 [−0.0270, 0.0291]
RAGTruth pooled	m3−m0	+0.0451 [0.0348, 0.0554]	+0.0962 [0.0644, 0.1295]

Note. HaluEval groups are item pairs (1,500 groups); RAGTruth QA has 150 source groups and the pooled split has 450. The QA AUROC interval includes zero, so the held-out transfer result is not resolved. The pooled row measures adaptation to task types seen in training, not unseen-task transfer.

5.5. Explanation, Efficiency, and Judges

The explanation results use the fixed-hyperparameter m3 checkpoint at seed 42. Figure 10 ranks the top-15 features by mean absolute SHAP over 1,000 stratified HaluEval test rows. Context overlap dominates at 1.66, word count follows at 1.11, question-answer cosine at 0.96, character count at 0.82, and question-answer Jaccard at 0.78. The claim-contradiction mean enters at rank eight with 0.22. The top ten features carry 77.9% of the total attribution mass. Exact TreeSHAP reproduces the model margin to 9.2×10^{-6}. The m3 ranking agrees with the frozen B2 ranking (Spearman $\rho = 0.878$, Kendall $\tau = 0.730$), so the added claim features refine the explanation rather than replace the original drivers. Attribution stability across seeds is high. Against seed 42, the global mean absolute SHAP ranking has Spearman $\rho = 0.972$ for seed 123 and 0.950 for seed 456, the top-10 overlap is nine of ten, and the top-5 set is identical for all three seeds (context overlap, word count, question-answer cosine, character count, question-answer Jaccard). The explanation is therefore stable to the seed, although the perturbation pilot below shows that it is less stable to content edits. The attributions are on the raw margin, not on the display score. The audit cases are error-enriched, and their roles change under m3; they are discussed below.

Table 14 shows the m3 pilot. Content edits move the score: entity replacement 0.339, support removal 0.464, number replacement 0.332, and date replacement 0.482. Structural edits barely move it: irrelevant insert 0.003 and clause shuffle 0.0001. The top-1 feature flips in 54.5% of entity perturbations and in none of the other operators, and Spearman correlation stays between 0.73 and 0.999. The entity flip rate is higher than in the B2 run, so the m3 explanation is less stable when the answer entity changes. The pilot has between two and twenty changed cases per operator, so these values are descriptive, not effect estimates.

A structured AI-assisted audit assessed 26 cases that were selected under B2 and re-scored under m3. Under m3 the same cases redistribute to 7 false positives, 12 false negatives, 3 borderline cases, 3 other cases, and 1 true positive. The labels are a draft for author verification, not independent human annotation, so the counts below are provisional. Each explanation was checked against the answer-context evidence, not the dataset label. Thirteen of the 26 explanations were judged implausible, three unsure, and ten plausible. The explanations are diagnostic aids, not faithful attributions. False positives inherit the length confound, and false negatives often reuse entities from the context. Two cases (2251

Table 14
Perturbation stability of the m3 raw score and SHAP ranking on HaluEval (seed 42, 20-case pilot).

Operator	n	Δscore	Large rate	Top-1 flip	Spearman
Entity replacement	11	0.339	0.364	0.545	0.848
Support removal	10	0.464	0.500	0.000	0.808
Number replacement	3	0.332	0.333	0.000	0.731
Date replacement	2	0.482	0.500	0.000	0.786
Irrelevant insert	20	0.003	0.000	0.000	0.922
Clause shuffle	2	0.000	0.000	0.000	0.999

Note. n counts perturbations that changed the feature vector. The pilot covers 20 cases and 120 attempts, of which 48 changed. Per-operator n is small, so the deltas are descriptive. Δscore is the mean absolute change in the m3 raw score.

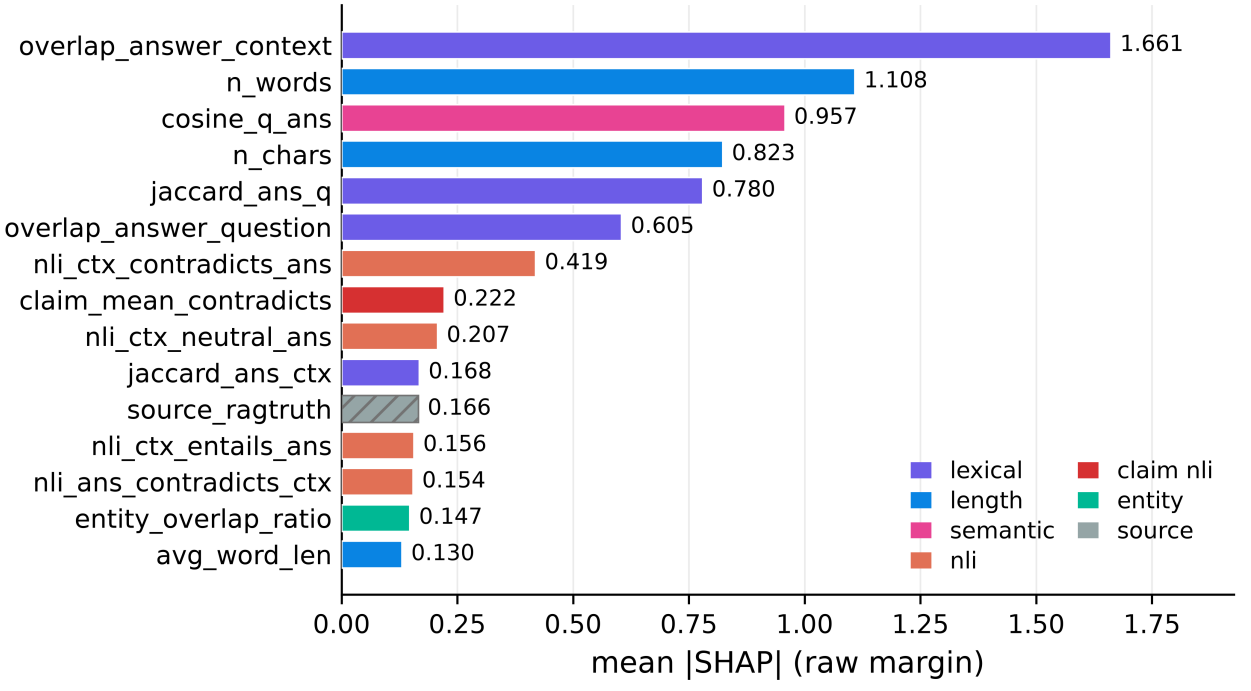


Figure 10: Mean absolute TreeSHAP for the fixed-hyperparameter EC-XGB m3 model, top-15 features (seed 42, 1,000 test rows). Hatching marks a feature that is constant on the sample; its attribution arises through interactions.

and 2436) appear to be label noise. Because the sample is error-enriched, the percentages describe the audit, not the test set. A three-author blinded review with AI pre-annotation of this sheet and a 60-item overlap is the final validation step, so no human kappa is reported here.

Table 15 gives the component latency. NLI (27.1 ms) and NER (21.1 ms) dominate the base median. XGBoost is 1.26 ms and SHAP 1.9 ms. The 1.26 ms entry is the legacy B2 model. EC-XGB model-only inference is 0.008 ms, and the 61.8 ms figure is the base 26-feature pipeline. Adding the eight claim features raises the m3 path to 127.5 ms at the median, with a p95 of 234.6 ms; claim NLI alone is 43.8 ms. The artifact is 0.68 MB, and the marginal cost is about $0.001 per 1,000 predictions. The same 200-row protocol on CPU takes 2.00 s per answer on average in a single contended run (p50 0.98 s, p95 7.56 s), so the 127.5 ms figure depends on GPU feature extraction; the classifier and SHAP steps remain negligible. Feature extraction, not the model, is the cost. Table 16 compares cost and latency: classical heads run in under a millisecond, the m3 pipeline is 127.5 ms, and the two judges are 18 to 24 times slower. The heuristic needs no model and costs almost nothing. RoBERTa-base inference on the same 200 rows takes 6.97 ms per row at batch 8 and fp16, so the m3 feature pipeline is about 18 times slower than the encoder alone. The encoder still needs a fine-tuned classification head, and its training cost is not included.

Table 15
Per-component latency in milliseconds (n = 200).

Component	p50	p95	mean
Core lexical	0.10	0.15	0.11
Entity (spaCy NER)	21.13	47.24	25.34
NLI (2 directions)	27.09	68.60	40.09
Semantic (MiniLM)	9.25	15.55	12.56
XGBoost inference	1.26	8.80	2.62
SHAP explanation	1.91	15.52	3.65
Base total	61.77	—	—
Claim NLI (m3)	43.75	83.21	53.11
Total m3	127.51	234.59	145.89

Note. Components are timed in separate instrumented passes over the same samples, so the component means need not sum to the end-to-end median. The base total is the 26-feature pipeline; the m3 total adds the eight claim-level NLI features. Retrieval and judge calls are excluded.

Table 16
Per-baseline cost and latency.

Baseline	Inputs	Size	p50 (ms)	Cost/1K
Heuristic (overlap)	text	—	0.009	≈$0
Logistic regression	26 feats	1.1 KB	0.001	≈$0.001
Random forest	26 feats	8.4 MB	0.331	≈$0.001
NLI-only	6 NLI probs	—	—	≈$0.001
TF-IDF (answer/full)	text	0.58/4.1 MB	0.014/0.110*	≈$0.001
XGBoost (conventional)	26 feats	0.49 MB	0.005	≈$0.001
RoBERTa-base	256 tokens	125M params	6.97	training only
EC-XGB (m3)	35 feats	0.68 MB	127.51	≈$0.001
GPT 5.6 Luna judge	API	—	2283.07	$0.234
Qwen 3.5 4B judge	4-bit local	4B params	3049.96	$0

Note. Size is the seed-42 artifact on disk. Feature models share the 127.5 ms m3 pipeline end to end; the model-only inference time is 0.008 ms. The NLI-only head was not persisted, so its cost is the 27.1 ms NLI step plus a linear head of the same order as logistic regression. * marks vectorizer-transform-only timings. RoBERTa timing is encoder-only (batch 8, fp16, 512 tokens, RTX 3060) and excludes the classification head and training. Costs are USD per 1,000 predictions.

LLM judges are not fair evaluators, and their preferences can shift with answer order [40]. Table 17 compares EC-XGB with two judges on the 900 RAGTruth QA rows at 17.8% prevalence. GPT-5.6 Luna reaches an accuracy of 0.743, an F1 of 0.565, an AUROC of 0.910, and flags 41.4% of rows; Qwen 3.5 4B reaches an accuracy of 0.850, an F1 of 0.602, an AUROC of 0.778, and flags 19.9%. EC-XGB reaches an F1 of 0.310, an AUROC of 0.558, and flags 96.8%. Both judges outperform EC-XGB on F1 and AUROC, and the target-fitted length-only baseline at AUROC 0.712 also ranks the QA rows better than the full feature model. GPT costs $0.234 per 1,000 calls and takes 2.28 s; Qwen is local and takes 3.05 s. The feature model runs in 127.5 ms and costs about $0.001, but it is not the better QA detector. Under matched multi-source training, the adapted encoder reaches 0.694 on the seen RAGTruth tasks, above EC-XGB's 0.582, so the encoder is the stronger model on the tasks both have seen. The feature model is less degraded on unseen tasks, but both are near chance. The length-matched panel changes the ranking. On the 158-row exact-length HaluEval set, EC-XGB reaches F1 0.907 and AUROC 0.956, GPT reaches F1 0.865 and AUROC 0.957, and Qwen reaches F1 0.821 and AUROC 0.863. The feature model therefore matches the stronger judge on ranking, and its seed-mean F1 is 0.042 higher, but the paired item-bootstrap interval for GPT minus EC-XGB is −0.052 [−0.127, 0.024], which includes zero; the AUROC interval is +0.008 [−0.034, 0.048]. The natural-prevalence QA row is a domain-shift result, not a general judge ranking. Prompt choice also moves the judge. On 100 HaluEval rows, GPT's F1 ranges from 0.814 to 0.854 and AUROC from 0.910 to 0.954 across the current prompt, a strict evidence-only

prompt, and a concise JSON prompt; the strict prompt is best, but the range is about the size of the model gap on this benchmark.

Table 17
Judge stress tests: two judges, EC-XGB, length-only, and all-positive references on the 900 RAGTruth QA test rows at natural prevalence (17.8%), and on the 158-row length-matched HaluEval set (50% positive).

Model	P	R	F1	AUROC	Flagged	p50 (ms)	Cost/1K
GPT 5.6 Luna judge	0.404	0.942	0.565	0.910	41.4%	2283	$0.234
Qwen 3.5 4B judge	0.570	0.638	0.602	0.778	19.9%	3050	$0
EC-XGB (m3)	0.183	0.998	0.310	0.558	96.8%	127.5	≈$0.001
Length-only	0.439	0.181	0.257	0.712	7.3%	—	≈$0
All-positive	0.178	1.000	0.302	—	100.0%	—	≈$0
Length-matched HaluEval, 158 rows, 50% positive							
GPT 5.6 Luna judge	0.945	0.798	0.865	0.957	42.2%	2283	$0.234
Qwen 3.5 4B judge	1.000	0.696	0.821	0.863	34.8%	3050	$0
EC-XGB (m3)	0.954	0.865	0.907	0.956	45.4%	127.5	≈$0.001
Length-only	0.515	0.443	0.476	0.521	43.0%	—	≈$0

Note. The upper panel is the 900 RAGTruth QA test rows at 17.8% prevalence. GPT uses default decoding and seeds 42/123/456, mean over three repeats; 2,699 of 2,700 calls parsed, and one empty response was scored as grounded at 0.5. Qwen uses local 4-bit decoding with one seed; it is deterministic at temperature 0. EC-XGB is the seed-mean fixed-hp m3 configuration at the 0.5 threshold; the deployed tuned checkpoint gives AUROC 0.549 and a 97.4% flag rate. The lower panel is the 158-row exact-length matched HaluEval set (79 pairs) from the length control; GPT is the mean over three seeds, Qwen is one deterministic seed, and EC-XGB is the three-seed mean. The lower panel controls length but not domain, so it is a stress test, not a second natural-prevalence estimate. Cost/1K is USD at the recorded token counts and an assumed $1.20/M output price.

5.6. Error Analysis and Discussion

On the seed-42 fixed-hyperparameter run, m3 makes 43 errors on the HaluEval test set: 8 false positives and 35 false negatives, an error rate of 1.4%. The WP5 audit covers all eight false positives and 12 of the 35 false negatives, chosen to cover the codebook rather than sampled at random. The 20-case draft assigns 13 content errors and seven audit artifacts. The artifact group is dominated by grounded answers that the detector over-flags (four cases) and one-word answers with zero lexical overlap (two cases); one case is benchmark label noise. The content group is dominated by entity errors (conflation, distortion, misattribution, and fabricated names), followed by generic or descriptive answers, coreference, numeric or temporal errors, relation inversion, and wrong granularity. These counts describe the reviewed cases, not the 3,000-row test population, and the labels are an author-verification draft rather than independent human annotation. A separate 50-case pre-annotation sheet covers the eight threshold false positives, the 17 highest-scoring negatives, and 25 false negatives; its AI-suggested counts are 30 artifact cases (21 grounded over-flags, 7 label-noise cases, 2 short-answer over-flags) and 20 content cases across 12 categories. It is the input to the three-author blinded review with AI pre-annotation, not a human result. Table 18 shows why length matters for the conventional model. The hallucination rate rises from 2.1% for one-word answers to 12.7% at three to four words and 90.2% at five to eight words, then to 99.2% at 17+ words. False positives concentrate in the five to eight bin, and false negatives in the short bins. Answer length is therefore a near-perfect label indicator on HaluEval. The mean score follows the same curve, from 0.024 to 0.993.

Four controls test the length explanation. A logistic regression on the two length variables reaches F1 0.9457 and AUROC 0.9759. A reweighting control equalizes six length bins with 247 examples each. On that set, EC-XGB reaches F1 0.9867 and AUROC 0.9987, against 0.9475 and 0.9758 for a length-only baseline. The gain is +0.039 F1 and +0.023 AUROC, so the length cue is not the whole story. The control does not prove the shortcut is gone.

A length-matching control pairs one positive and one negative at each length, giving 79 pairs (158 examples) over 1,000 bootstraps. EC-XGB holds AUROC 0.956 while the length-only baseline falls to 0.521, a gap of 0.436 [0.351, 0.522]. The F1 gap is 0.440 [0.334, 0.552]. The length cue explains the zero-shot failure but not the in-domain ranking.

Figure 11 shows the same pattern: the rate jumps from 12.7% at three to four words to 90.2% at five to eight words, and false negatives are most frequent by rate in the short bins; by count, 18 of 36 are in the five to eight

Table 18
HaluEval test errors of the conventional B2 model by answer length (counts seed 42, rates three-seed means).

Words	Rows	Hallucinated rate	FP rate	FN rate	Mean score
1	524	0.021	0.004	0.182	0.024
2	514	0.033	0.000	0.176	0.035
3–4	474	0.127	0.002	0.178	0.126
5–8	641	0.902	0.106	0.031	0.888
9–16	600	0.982	0.000	0.003	0.980
17+	247	0.992	0.333	0.000	0.993

Note. Counts are seed 42. Rates and mean scores are three-seed means. The 17+ bin contains only two negative rows. The 5–8 word bin contains 66.7% of the false positives and 18 of the 36 false negatives.

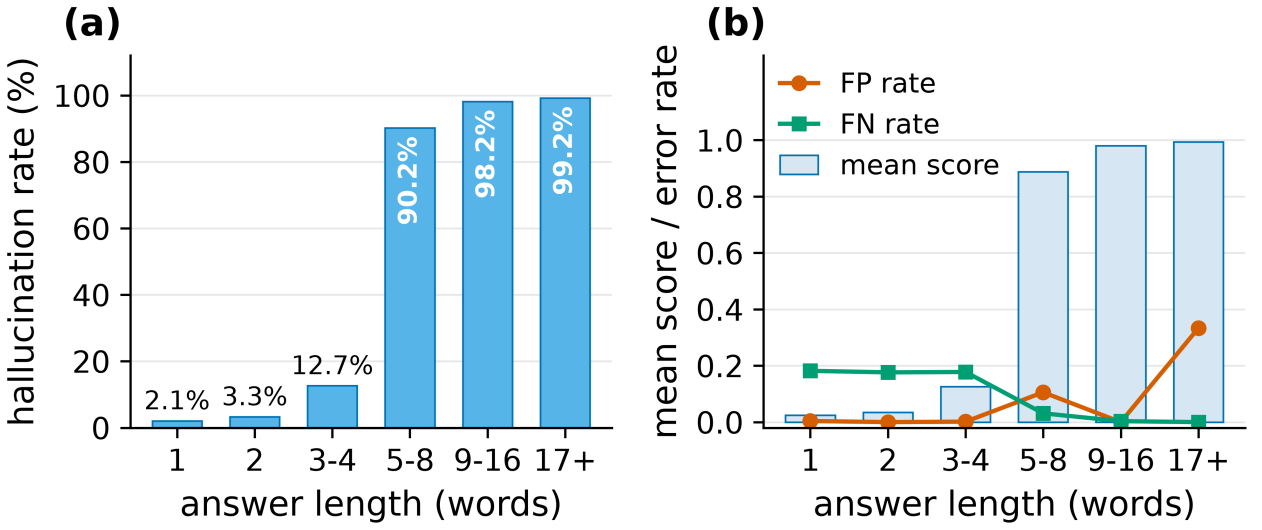


Figure 11: Hallucination rate, mean score, and error rates by answer length for the B2 conventional model.

bin. The shape explains the zero-shot result: complete sentences are long, so the conventional model reads them as hallucinations. The informative results are outside the benchmark. EC-XGB ranks the RAGTruth task types seen in training better than the conventional model, and the display calibrator gives readable target scores. The gain comes from multi-source training, not a different learning system. HaluEval labels length: 2.1% of one-word answers are hallucinations, and 99.2% at 17+ words. The ablation leaves lexical and length as the only groups that matter, so the in-domain model is dominantly a length and overlap detector. That is why natural long answers are flagged at 99.9%, calibration cannot repair the ranking, and held-out QA stays near 0.55. The contribution is the protocol that exposes this chain, not a general detector. A fine-tuned encoder still wins in-domain, so the choice is a cost-transparency-accuracy trade-off. The external comparison is limited by protocol. No published HaluEval QA result uses the grouped split of this study. A supervised 13B agent reports F1 0.724 on 100 HaluEval QA rows [8], and the benchmark's zero-shot judges report 62.6% to 69.8% accuracy [14]. Those settings differ in supervision, evidence access and test rows, so they are context rather than a head-to-head bar. The benchmark's generation instruction asks for hallucinated answers about five words longer than correct ones; the observed gap is 8.5 words. In this test split the mean answer length is 2.28 words for faithful answers and 10.79 for hallucinated answers, and a length-only logistic regression reaches AUROC 0.976 (Table 5). Any comparison that ignores this construction artifact overstates semantic detection. Five limits bound these claims. Web-search retrieval quality and end-to-end verdict accuracy are not evaluated; the retrieval stress test uses the known source passage. The judge comparison uses 900 RAGTruth QA rows at natural prevalence and the 158-row length-matched set, which controls length but not domain. The audit is error-enriched. The HaluEval benchmark is saturated, so its in-domain differences are not resolvable. The deployed display calibrator was selected

in an earlier fit that compared Platt and isotonic on the test split; the calibration analysis in Table 8 refits Platt without any test-based selection.

6. Conclusion

A useful hallucination detector must be fast, inexpensive, and explainable. This research has created EC-XGB, an evidence-consistent XGBoost risk predictor. EC-XGB uses 26 evidence-consistency features, 8 claim-level NLI aggregates, and a source indicator, with log-scaled lengths under monotone constraints, multi-source training, calibrated display scores, and SHAP explanations. On HaluEval QA the fixed-hp model reaches an F1 of 0.9854 and an AUROC of 0.9986, and the tuned deployed checkpoint is 0.9855 and 0.9984, which matches conventional XGBoost on a saturated benchmark. Under a single fixed configuration, the four variants span F1 0.9849 to 0.9854, so the eight claim aggregates add no aggregate F1 gain. The fixed-hp m3 feature ablation shows that lexical overlap and length carry the in-domain score: removing them costs 0.0171 and 0.0066 F1, and every other group changes F1 by less than 0.0041. Across seeds, the global SHAP ranking is stable (Spearman 0.950 to 0.972; top-5 identical), so the explanation is reproducible even though it is dominated by length and overlap.

On the RAGTruth tasks used in training, the pooled flag rate falls from 99.9% to 61.3%, while held-out QA remains at 96.8% and AUROC 0.558. Target calibration lowers ECE from 0.731 to 0.125, but a constant predictor at the test prevalence has a lower Brier score, and the calibrated score flags no target rows at 0.5; on the conventional reference, a 5% false-positive budget recalls 5.0%. Split conformal covers 94.4% at a 90% target on QA, but positive coverage is only 0.685; a class-conditional quantile restores it to 0.887, and FaithBench coverage falls to 0.320. On the independent MedHallu corpus the model ranks the paired hallucinated answer above the grounded one in 86.7% of 1,000 pairs (AUROC 0.821), but flags every row and is badly calibrated (ECE 0.493), so the ranking survives the domain change while the probability does not. On HaluEval dialogue and summarization the model collapses: dialogue equals the all-positive baseline, and summarization is below the length-only ranking (AUROC 0.717 vs 0.788). Under matched multi-source training, the adapted encoder reaches 0.694 on the seen RAGTruth tasks, above EC-XGB's 0.582, while the source-only encoder wins in-domain (0.9912 F1) but is near chance on each task (0.458 and 0.491). On unseen QA both are near chance (AUROC 0.490 vs 0.549). One analysis takes 127.5 ms on GPU and 2.00 s on CPU, and costs about $0.001 per 1,000 predictions. The GPT 5.6 Luna judge reaches F1 0.565 and AUROC 0.910 on the natural QA rows, but on the 158-row length-matched set EC-XGB reaches F1 0.907 and the judge 0.865. EC-XGB is therefore a domain-specific risk assessment tool and adaptation check, not a general-purpose detector. Future work will cover domain adaptation for medical and multi-task QA, stronger multicalibration, and threshold transfer.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Saiful Alam Sabbir:** Writing - review & editing. **MD. Miraz Ahamed:** Writing - review & editing. **Ohidujjaman Tuhin:** Supervision, Validation, Writing - review & editing.

Declarations

- **Funding:** No external funding was provided to conduct this research.
- **Conflicts of Interest:** The researchers declare no conflict of interest.

- **Data availability:** Raw evaluation datasets are not redistributed. The repository includes download scripts, source revisions, SHA-256 hashes, and license notes for HaluEval (MIT), RAGTruth (MIT), FaithBench (CC BY-NC-SA 4.0), MedHallu (MIT), and the HaluBench subset (CC BY-NC 2.0). Splits and feature matrices can be reproduced from the original dataset sources. Model artifacts, split indices, and the experiment manifest are included in the release archive.
- **Code availability:** The training pipeline, FastAPI inference service, and Next.js dashboard are available in the project repository.
- **Ethical approval:** This research employs publicly available benchmark datasets on their existing licensing terms. The research does not involve human subjects, and hence no ethics approval is needed.

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References

[1] P. Manakul, A. Liusie, M. J. Gales, Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models, in: Proceedings of EMNLP, 2023, pp. 9004–9017. doi:10.18653/v1/2023.emnlp-main.557.

[2] M. Belyi, R. Friel, S. Shao, A. Sanyal, Luna: A lightweight evaluation model to catch language model hallucinations with high accuracy and low cost, in: Proceedings of COLING (Industry Track), 2025, pp. 398–409.
URL https://aclanthology.org/2025.coling-industry.34/

[3] S. Farquhar, J. Kossen, L. Kuhn, Y. Gal, Detecting hallucinations in large language models using semantic entropy, Nature 630 (8017) (2024) 625–630. doi:10.1038/s41586-024-07421-0.

[4] S. Yadav, N. K. Verma, A hybrid framework for hallucination detection in large language models, IEEE Transactions on Artificial Intelligence (2026). doi:10.1109/TAI.2026.3653354.

[5] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, ACM Computing Surveys 55 (12) (2023) 1–38. doi:10.1145/3571730.

[6] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, T. Liu, A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions, ACM Transactions on Information Systems 43 (2) (2025) 1–55. doi:10.1145/3703155.

[7] A. Alansari, H. Luqman, Large language models hallucination: A comprehensive survey, Computer Science Review 61 (2026) 100970. doi:10.1016/j.cosrev.2026.100970.

[8] X. Cheng, J. Li, W. X. Zhao, H. Zhang, F. Zhang, F. Zhang, K. Gai, J.-R. Wen, Small agent can also rock! empowering small language models as hallucination detector, in: Proceedings of EMNLP, 2024, pp. 14600–14615. doi:10.18653/v1/2024.emnlp-main.809.

[9] S. Valentin, J. Fu, G. Detommaso, S. Xu, G. Zappella, B. Wang, Cost-effective hallucination detection for llms, arXiv preprint arXiv:2407.21424 (2024).

[10] D. Sundaragiri, L. M. Reddy, P. Gunavardhan, B. Navahith, Framework for hallucination detection in large language models, International Journal of Engineering Research & Technology (IJERT) 15 (04) (2026). doi:10.5281/zenodo.20025987.

[11] I. Haq, M. Saqib, Y. Zhang, I. A. Khan, Quantifying factual divergence in generative models: Shap-lime based hallucination score for llms, Multimedia Systems 32 (2) (2026). doi:10.1007/s00530-025-02150-4.

[12] Y. Deng, Z. Fang, S. Li, L. Chen, Beyond in-domain detection: Spikescore for cross-domain hallucination detection, in: Proceedings of ICLR, 2026.

[13] F. S. Bao, M. Li, R. Qu, et al., Faithbench: A diverse hallucination benchmark for summarization by modern llms, in: Proceedings of NAACL (Short Papers), 2025, pp. 448–461. doi:10.18653/v1/2025.naacl-short.38.

[14] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, J.-R. Wen, Halueval: A large-scale hallucination evaluation benchmark for large language models, in: Proceedings of EMNLP, 2023. doi:10.18653/v1/2023.emnlp-main.397.

[15] C. Niu, Y. Wu, J. Zhu, S. Xu, K. Shum, R. Zhong, J. Song, T. Zhang, Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models, in: Proceedings of ACL, 2024, pp. 10862–10878. doi:10.18653/v1/2024.acl-long.585.

[16] S. Lin, J. Hilton, O. Evans, Truthfulqa: Measuring how models mimic human falsehoods, in: Proceedings of ACL, 2022, pp. 3214–3252. doi:10.18653/v1/2022.acl-long.229.

[17] S. Min, K. Krishna, X. Lyu, M. Lewis, W.-t. Yih, P. W. Koh, M. Iyyer, L. Zettlemoyer, H. Hajishirzi, Factscore: Fine-grained atomic evaluation of factual precision in long form text generation, in: Proceedings of EMNLP, 2023, pp. 12076–12100. doi:10.18653/v1/2023.emnlp-main.741.

[18] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of ICML, 2017.
URL http://proceedings.mlr.press/v70/guo17a.html

[19] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems (NeurIPS), 2017.
URL https://papers.nips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[20] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD, 2016. doi:10.1145/2939672.2939785.

[21] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, J. Snoek, Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift, in: Advances in Neural Information Processing Systems (NeurIPS), 2019.

[22] P. He, J. Gao, W. Chen, DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing, in: Proceedings of the International Conference on Learning Representations (ICLR), 2023.

[23] S. R. Bowman, G. Angeli, C. Potts, C. D. Manning, A large annotated corpus for learning natural language inference, in: Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2015, pp. 632–642. doi:10.18653/v1/D15-1075.

[24] A. Williams, N. Nangia, S. R. Bowman, A broad-coverage challenge corpus for sentence understanding through inference, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2018, pp. 1112–1122. doi:10.18653/v1/N18-1101.

[25] N. Reimers, I. Gurevych, Sentence-bert: Sentence embeddings using siamese bert-networks, in: Proceedings of EMNLP-IJCNLP, 2019, pp. 3982–3992. doi:10.18653/v1/D19-1410.

[26] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, M. Zhou, MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers, in: Advances in Neural Information Processing Systems (NeurIPS), 2020.

[27] M. Honnibal, I. Montani, S. Van Landeghem, A. Boyd, spaCy: Industrial-strength natural language processing in Python (2020). doi:10.5281/zenodo.1212303.

[28] D. Ru, L. Qiu, X. Hu, T. Zhang, P. Shi, S. Chang, J. Cheng, C. Wang, S. Sun, H. Li, Z. Zhang, B. Wang, J. Jiang, T. He, Z. Wang, P. Liu, Y. Zhang, Z. Zhang, RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 37, 2024, pp. 21999–22027. doi:10.52202/079017-0692.

[29] J. C. Platt, Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods, in: A. J. Smola, P. Bartlett, B. Schölkopf, D. Schuurmans (Eds.), Advances in Large Margin Classifiers, MIT Press, Cambridge, MA, 1999, pp. 61–74.

[30] B. Zadrozny, C. Elkan, Transforming classifier scores into accurate multiclass probability estimates, in: Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2002, pp. 694–699. doi:10.1145/775047.775151.

[31] A. Gasparin, G. Detommaso, Distance-aware calibration for pre-trained language models, in: Findings of the Association for Computational Linguistics: EMNLP 2024, 2024, pp. 12434–12447. doi:10.18653/v1/2024.findings-emnlp.725.

[32] G. Kelodjou, L. Rozé, V. Masson, L. Galárraga, R. Gaudel, M. Tchuente, A. Termier, Shaping up SHAP: Enhancing stability through layer-wise neighbor selection, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38, 2024, pp. 13094–13103. doi:10.1609/aaai.v38i12.29208.

[33] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, B. Kim, Sanity checks for saliency maps, in: Advances in Neural Information Processing Systems (NeurIPS), 2018.

[34] Y. Bang, Z. Ji, A. Schelten, A. Hartshorn, T. Fowler, C. Zhang, N. Cancedda, P. Fung, HalluLens: LLM hallucination benchmark, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vienna, Austria, 2025, pp. 24128–24156. doi:10.18653/v1/2025.acl-long.1176.

[35] S. Pandit, J. Xu, J. Hong, Z. Wang, T. Chen, K. Xu, Y. Ding, MedHallu: A comprehensive benchmark for detecting medical hallucinations in large language models (2025). arXiv:2502.14302, doi:10.48550/arXiv.2502.14302.

[36] S. S. Ravi, B. Mielczarek, A. Kannappan, D. Kiela, R. Qian, Lynx: An open source hallucination evaluation model (2024). arXiv:2407.08488, doi:10.48550/arXiv.2407.08488.

[37] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, et al., RoBERTa: A robustly optimized BERT pretraining approach, arXiv preprint arXiv:1907.11692 (2019). doi:10.48550/arXiv.1907.11692.

[38] C. Chen, K. Liu, Z. Chen, Y. Gu, Y. Wu, M. Tao, Z. Fu, J. Ye, INSIDE: LLMs' internal states retain the power of hallucination detection, in: International Conference on Learning Representations (ICLR), 2024.

[39] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, I. Stoica, Judging LLM-as-a-judge with MT-Bench and Chatbot Arena, in: Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2023.

[40] P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, L. Kong, Q. Liu, T. Liu, Z. Sui, Large language models are not fair evaluators, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2024, pp. 9440–9450. doi:10.18653/v1/2024.acl-long.511.